

Hybrid Physics–ML Framework

Updated 4 October 2025

Hybrid physics–ML frameworks systematically integrate mechanistic models and data-driven corrections to leverage established physical laws and high-capacity learning.

They improve prediction accuracy and robustness, demonstrated by lower error metrics and reliable uncertainty quantification in domains like thermal engineering and climate science.

These frameworks are applied in diverse areas such as digital twins, computational physics, and energy systems, effectively addressing challenges like data scarcity and nonstationarity.

A hybrid physics–ML framework designates any modeling approach where a physics-based model and machine learning techniques are systematically integrated—rather than simply operated in parallel—such that their reciprocal strengths are combined. The primary objective is to leverage established physical understanding and mechanistic constraints alongside the data-adaptive, high-capacity pattern recognition capabilities offered by ML models. Hybrid frameworks have matured to address prediction, uncertainty quantification, generalization, and operational deployment challenges across domains including thermal engineering, dynamical systems, material science, climate science, [digital twins](#), and computational physics.

1. Fundamental Principles and Mathematical Structure

The archetype of a hybrid physics–ML framework is the “gray-box” model, in which a physics-based surrogate is [augmented](#) by a data-driven correction. The typical mathematical form for many applications is

$$\hat{y} = \hat{y}_p + \hat{\epsilon}_m$$

where $\hat{y}_p$ is the prediction from a domain-knowledge (DK) model (for instance, an empirical correlation or a reduced physical theory), and $\hat{\epsilon}_m$ is an ML-predicted residual trained on the mismatch between experimental observations y and $\hat{y}_p$ ([Zhao et al., 2019](#), [Kuberan et al., 28 Jan 2025](#), [Furlong et al., 26 Feb 2025](#)).

Alternate designs embed a physics-based model as a feature extractor within a neural architecture (e.g., the injection of outputs from a panel method into a hidden layer of a neural network) (Pawar et al., 2021, Pawar et al., 2021) or as a fully differentiable component that enables gradient-based co-optimization of physical and experimental parameters (Malik et al., 8 Aug 2025). Some frameworks employ game-theoretic or cooperative learning schemes where the physics and ML models act as independent but mutually regularized agents (Liverani et al., 17 Sep 2025).

Physics constraints can also be enforced directly on the ML posterior or loss function, for instance via penalties derived from partial differential equations (PDEs) or variational principles integrated in a Boltzmann-Gibbs regularizer for Gaussian Processes (Chang et al., 2022).

Architecturally, frameworks range from residual-corrected base models (correction networks), physics-guided neural networks (PGML, PINNs), multi-fidelity interfaces, to cooperative or modular systems in which the physical and ML modules interact through carefully constructed loss functions or mutual consistency criteria.

2. Predictive Capability and Generalization

Hybrid frameworks offer significant improvements in prediction accuracy and generalization, especially in regimes characterized by sparse data, high-dimensional feature spaces, or nonstationarity. The physics-based component provides a physically plausible scaffold, anchoring predictions to meaningful domains, while the ML component captures nontrivial corrections, trends, or unmodeled effects (Zhao et al., 2019, Kuberan et al., 28 Jan 2025, Espinel-Ríos et al., 2024). This synergy is most apparent in settings where the domain knowledge is incomplete or the governing equations are only approximate, and in data regimes where purely black-box ML models risk overfitting or catastrophic extrapolation errors (Furlong et al., 26 Feb 2025, Kriuk, 2 Oct 2025).

Empirically, hybrid models for critical heat flux (CHF) and nucleate pool boiling have achieved relative errors below 2% (mean absolute relative error $\approx 1.8\%$ with ensemble DNNs; $R^2 \approx 0.995$ for hybrid deep learning/correlation models), far surpassing standalone ML or base physics predictors (Furlong et al., 26 Feb 2025, Kuberan et al., 28 Jan 2025). Furthermore, hybrid models are robust against data scarcity, providing physically reasonable predictions in extrapolative regimes where pure ML models degrade catastrophically (Furlong et al., 26 Feb 2025, Kriuk, 2 Oct 2025).

Hybrid digital twins in engineering combine efficient reduced-order physical models (ROM) with physics-guided ML to reduce predictive uncertainty by up to 75% and enable seamless multi-fidelity integration ([Pawar et al., 2021](#)). Hybrid approaches in permafrost infrastructure risk deliver $R^2 = 0.980$ (over 2.9 million samples) while maintaining physical validity in climate change scenarios ([Kriuk, 2 Oct 2025](#)).

3. Model Construction and Methodological Innovations

There are several methodological archetypes in hybrid physics–ML framework construction:

Residual Correction ("Gray-Box"): A physics-based prior is fixed. The residual between observed and model-predicted outputs is regressed via ML. The final output is the sum of the prior and ML-corrected residual ([Zhao et al., 2019](#), [Kuberan et al., 28 Jan 2025](#), [Furlong et al., 26 Feb 2025](#)).

Physics-Guided Neural Networks (PGML): Physics-based features are injected into hidden layers to enforce latent structure consistency and reduce uncertainty ([Pawar et al., 2021](#), [Pawar et al., 2021](#)).

Sequential Data Assimilation: ML networks (e.g., LSTMs for closure) are combined with physics solvers, and errors suppressed via sequential filtering (e.g., ensemble Kalman filters) ([Pawar et al., 2021](#)).

Cooperative/Mutual Regularization (HYCO): Independent physical and ML models are co-trained, nudged by an interaction loss that enforces mutual agreement without hard constraints ([Liverani et al., 17 Sep 2025](#)).

Differentiable Simulation–Neural Module Coupling: Fully differentiable physics simulators are coupled to neural network modules in a unified gradient optimization pipeline (e.g., electron diffraction with learned experimental thickening) ([Malik et al., 8 Aug 2025](#)).

Game Theoretic Formulations: The optimization is recast as a two-player game, with one loss for the physical model and another for the synthetic network, yielding Nash equilibrium solutions ([Liverani et al., 17 Sep 2025](#)).

Uncertainty Quantification: Hybrid models with DNN ensembles, Bayesian neural networks, and deep Gaussian processes provide both point prediction and credibility intervals; uncertainty calibration is verified using calibration curves ([Furlong et al., 26 Feb 2025](#), [Chang et al., 2022](#)).

Hybrid frameworks are frequently modular, supporting the substitution of different ML components, physical models, or loss formulations to accommodate

problem-specific physics or data.

4. Application Domains and Exemplary Results

Hybrid physics–ML approaches have broad applicability:

Thermal and Energy Systems: Reliable CHF and heat transfer coefficient (HTC) predictions are achieved by correcting empirical correlations via DNNs or random-forest regressions, with interpretability enhanced using [SHAP](#) values to map feature importance and robustness ([Zhao et al., 2019](#), [Kubera et al., 28 Jan 2025](#)).

Climate and Geoscience: Pan-Arctic permafrost degradation is forecasted at massive scales with statistical ML models blended with physics-informed adjustment (e.g., -10 pp/°C). This enables operational risk mapping for infrastructure and engineering codes under rapid warming ([Kriuk, 2 Oct 2025](#)).

Battery Systems: Voltage prediction and parameter estimation in Li-ion batteries are improved via hybrid models coupling single-particle (SPMe) models and Gaussian Process-corrected residuals, with explicit treatment of uncertainty via discrepancy functions ([Fogelquist et al., 10 May 2025](#)).

Crowd Dynamics: Heterogeneous pedestrian behavior including pushing at bottlenecks is captured by integrating random-forest behavior classifiers (driven by neighborhood features and internal propensities) with multi-mode collision-free velocity models ([Xu et al., 2024](#)).

Marine and Vehicle Dynamics: Imperfect physical models of hydrodynamics are augmented by residual feedforward neural networks (enhanced via state-dependent trigonometric features) to improve maneuvering predictions under environmental disturbances ([Wang et al., 2024](#)).

High-Dimensional PDEs and Computational Physics: Physics-informed CNN/[PINN](#) hybrids (e.g., BridgeNet) and frameworks embedding geometric priors (symplectic, Roe flux, vortex) directly into ML architectures yield robust, long-term stable, and physically consistent solutions to complex, high-dimensional and hyperbolic PDEs ([Mirzabeigi et al., 4 Jun 2025](#), [Tong, 2024](#)).

Materials and Multiscale Mechanics: Hybrid encoders-modifying evolving constitutive parameters are integrated with physics-based decoders, combining data-driven flexibility with physical path-dependence and interpretability ([Rocha et al., 2023](#)).

Astrophysics and Spectroscopy: Black hole spin estimation frameworks leverage the Teukolsky formalism inside PINNs, yielding models with high

interpretability, generalizability, and outperformance on sensitivity benchmarks ([Menziltsidou, 27 Jul 2025](#)).

Digital Twins: Hybrid interface-learning schemes merge ROMs and FOMs through learnable digital interfaces (potentially LSTMs) for multi-fidelity system replications ([Pawar et al., 2021](#)).

5. Robustness, Flexibility, and Physical Interpretability

Hybrid frameworks are designed for robustness in extrapolation and regime transfer. Anchoring predictions to physics-based priors greatly reduces the likelihood of unphysical outputs—a critical requirement for safety-critical domains. The use of fixed or interpretable physics modules not only improves reliability but also facilitates diagnostic tools: for example, PGML approaches can return confidence scores indicating the reliability of predictions given unknown physics ([Pawar et al., 2021](#)).

Because the ML module typically addresses only residual error, model complexity and computational cost are reduced relative to pure ML approaches. This feature is advantageous for rapid retraining and on-the-fly adaptation to new operational domains or configurations (e.g., new channel geometries in boiling system CHF analysis) ([Zhao et al., 2019](#)).

6. Open Problems, Limitations, and Future Directions

Hybrid frameworks introduce several new challenges and open research directions:

Model Complexity and Training Overhead: Integrating physics models (particularly when embedding integrators or high-fidelity simulators) increases training times and may require small time steps and explicit attention to numerical stability ([Tong, 2024](#)).

Parameter Estimation under Uncertainty: Jointly calibrating hybrid models remains nontrivial, necessitating advances in discrepancy modeling and scalable likelihood optimization for large time series ([Fogelquist et al., 10 May 2025](#)).

Generalization to High-Dimension and Nonstationarity: Ensuring transferability and consistency under domain shift or nonstationary driving conditions (such as rapid Arctic warming) continues to motivate research into principled blending weights and adaptive, context-aware hybridization strategies ([Kriuk, 2 Oct 2025](#)).

Game-Theoretic and Cooperative Learning: Mutual regularization schemes (as in HYCO) raise theoretical questions about convergence, stability, and

robustness under adversarial or decentralized training ([Liverani et al., 17 Sep 2025](#)).

Uncertainty Quantification and Calibration: While many hybrid models provide uncertainty estimates, ensuring their statistical calibration (especially in low-data, extrapolative, or safety-critical regimes) remains an active area ([Furlong et al., 26 Feb 2025](#), [Chang et al., 2022](#)).

Extension to Multiphysics and Modular Scientific Discovery: There is ongoing development toward scale-bridging interface learning and modular hybridization supporting complex multiphysics phenomena, adaptivity to missing data, and coupling with privacy-aware or [federated training](#) ([Pawar et al., 2021](#), [Tong, 2024](#), [Liverani et al., 17 Sep 2025](#)).

A plausible implication is that as hybrid frameworks become standard, modeling workflows will move toward compositional, modular designs with automated selection and tuning of the physics/ML blend according to data availability, forecasting need, and physical regime.

7. Summary Table: Hybrid Physics–ML Frameworks Across Domains

Domain	Physics Component	ML Component	Integration Mode
Thermal Systems	Empirical/physical correlations (e.g., LUT, CHF)	DNN, Random Forest	Residual correction (gray-box) (Zhao et al., 2019 , Kuberan et al., 28 Jan 2025)
Digital Twins	ROM, CFD models , panel methods	DNN, LSTM, PGML	Feature injection, interface learning (Pawar et al., 2021)
Battery Systems	Single Particle Model (SPMe)	Gaussian Process Regression	Residual error ML, discrepancy function (Fogelquist et al., 10 May 2025)
Crowd Dynamics	Generalized collision-free velocity models	Random Forest (behavior classifier)	Mode selection with ML-driven physics update (Xu et al., 2024)
Permafrost Risk	Empirical permafrost sensitivity (–10pp/°C)	RF, Histogram GB, Elastic Net	Weighted ensemble (60% ML, 40% physics) (Kriuk, 2 Oct 2025)
Computational Physics	Differentiable Bloch wave solver	Neural network (ThicknessNN)	Embedded differentiable module (Malik et al., 8 Aug 2025)

Domain	Physics Component	ML Component	Integration Mode
Astrophysics	Teukolsky formalism (angular PDE)	Physics-Informed Neural Network (PINN)	PDE-embedded PINN (Menziltsidou, 27 Jul 2025)
PDE Modeling	Discretized solver (e.g. FDM, FEM)	Neural network, synthetic agent	Cooperative alternating regularization (Liverani et al., 17 Sep 2025)

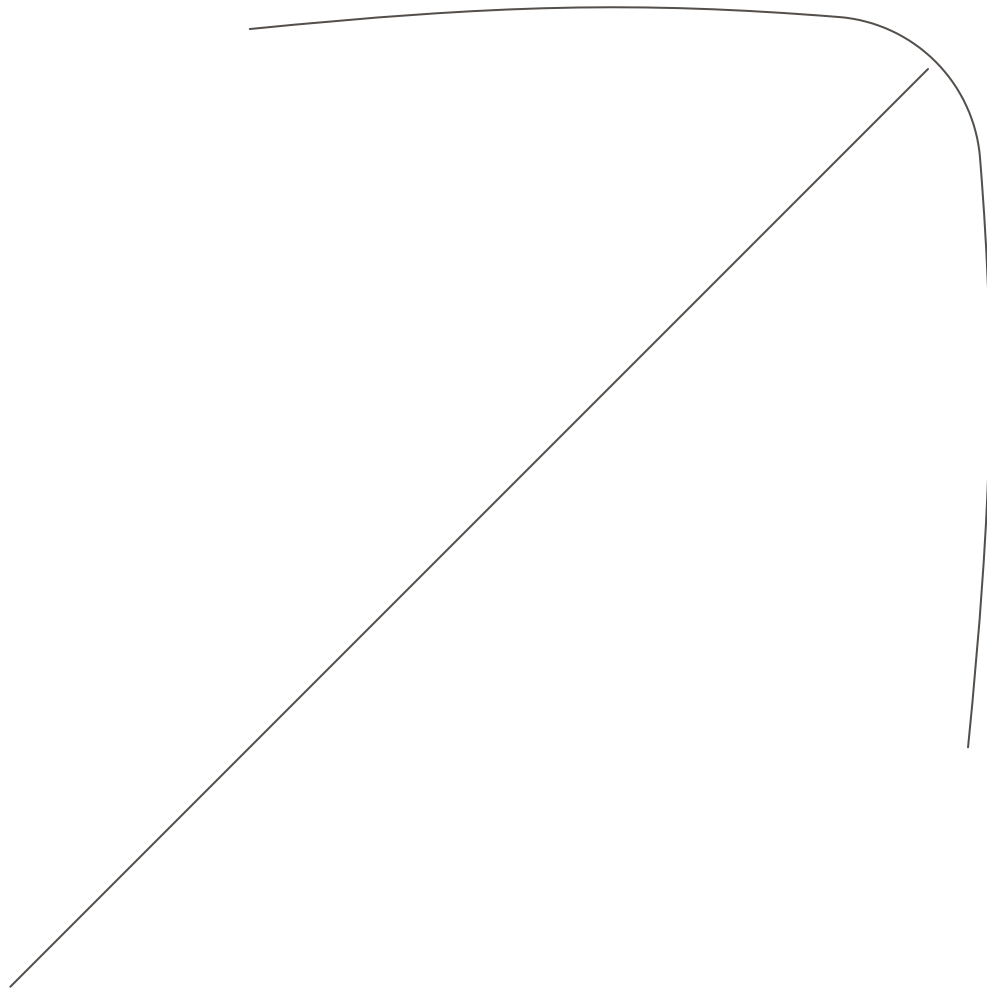
This summary table synthesizes representative frameworks and highlights characteristic integration strategies across key domains.

Hybrid physics–machine learning frameworks provide a systematically extensible methodology to unify mechanistic models and data-derived inference, producing robust, physically meaningful, and highly accurate models. Their adoption is increasing across scientific and engineering disciplines as they demonstrate unique advantages in generalization, uncertainty quantification, and operational reliability, especially in complex, data-limited, or extrapolative scenarios.

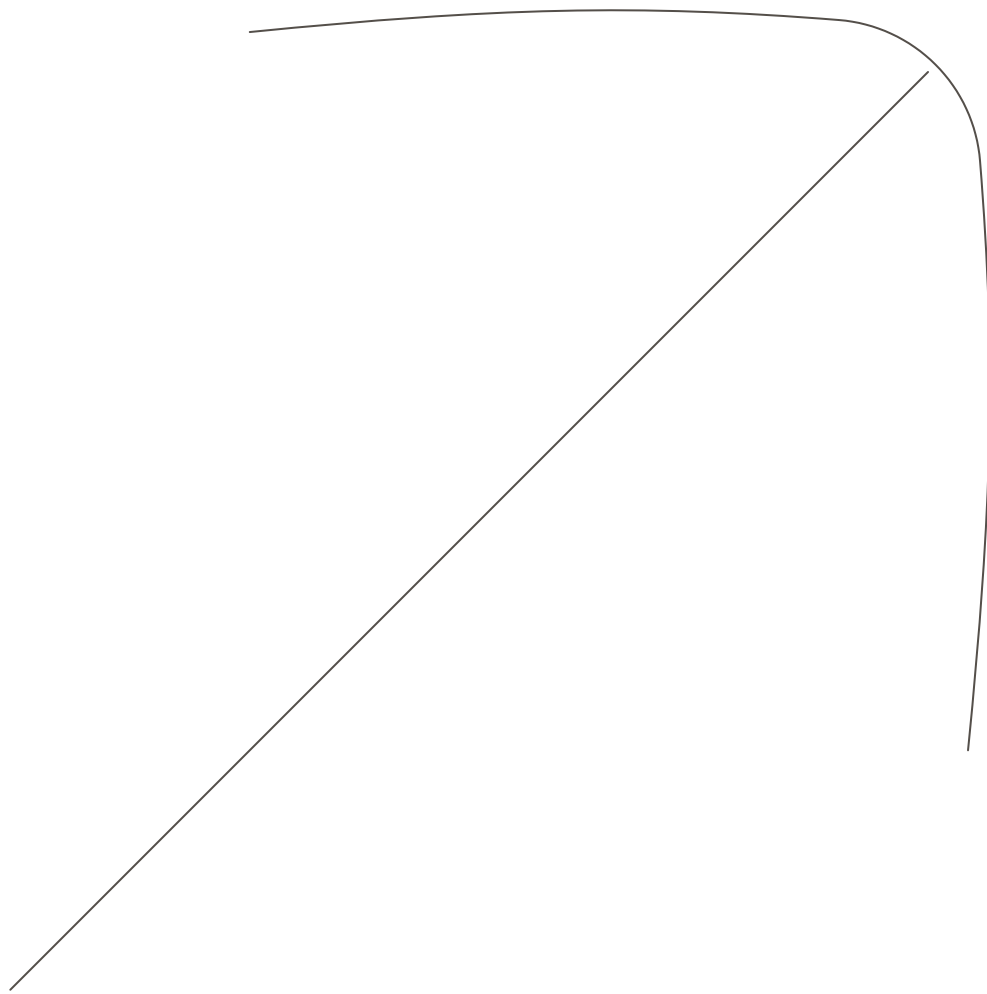




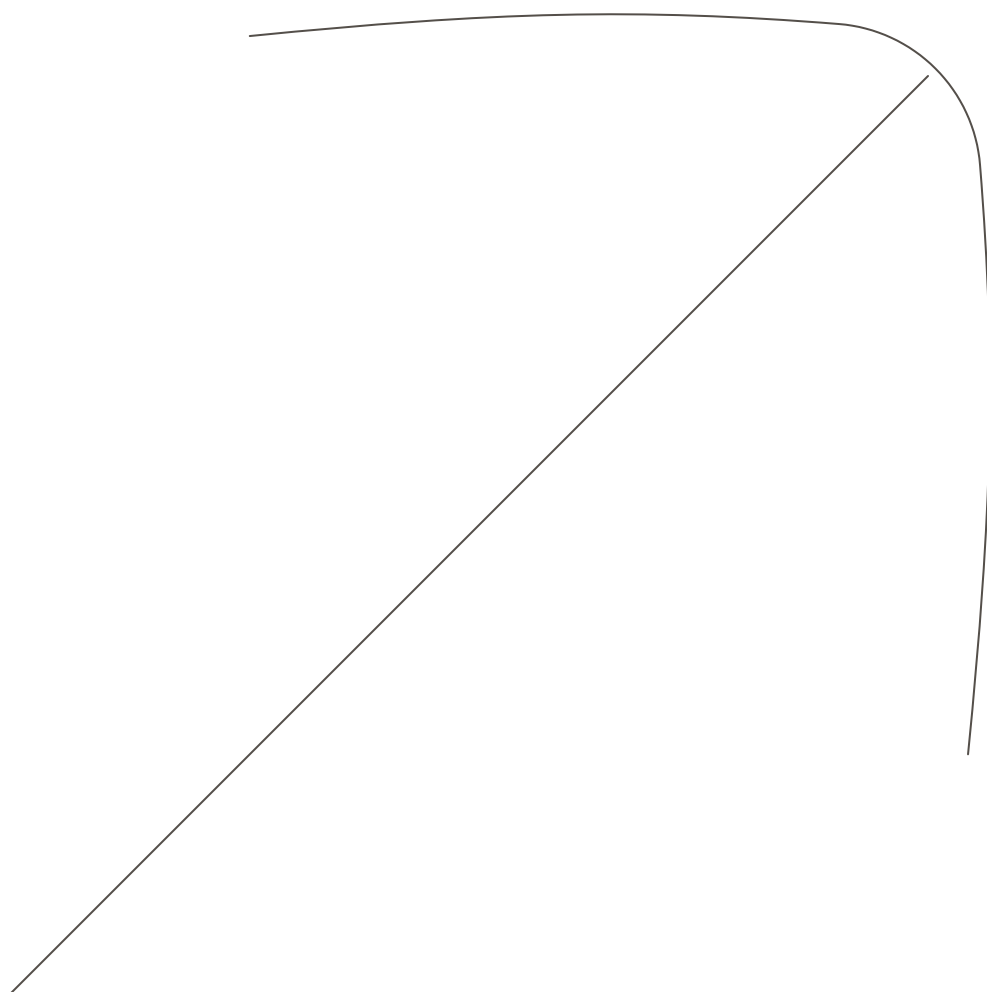
1. On the prediction of critical heat flux using a physics-informed machine learning-aided framework (2019)



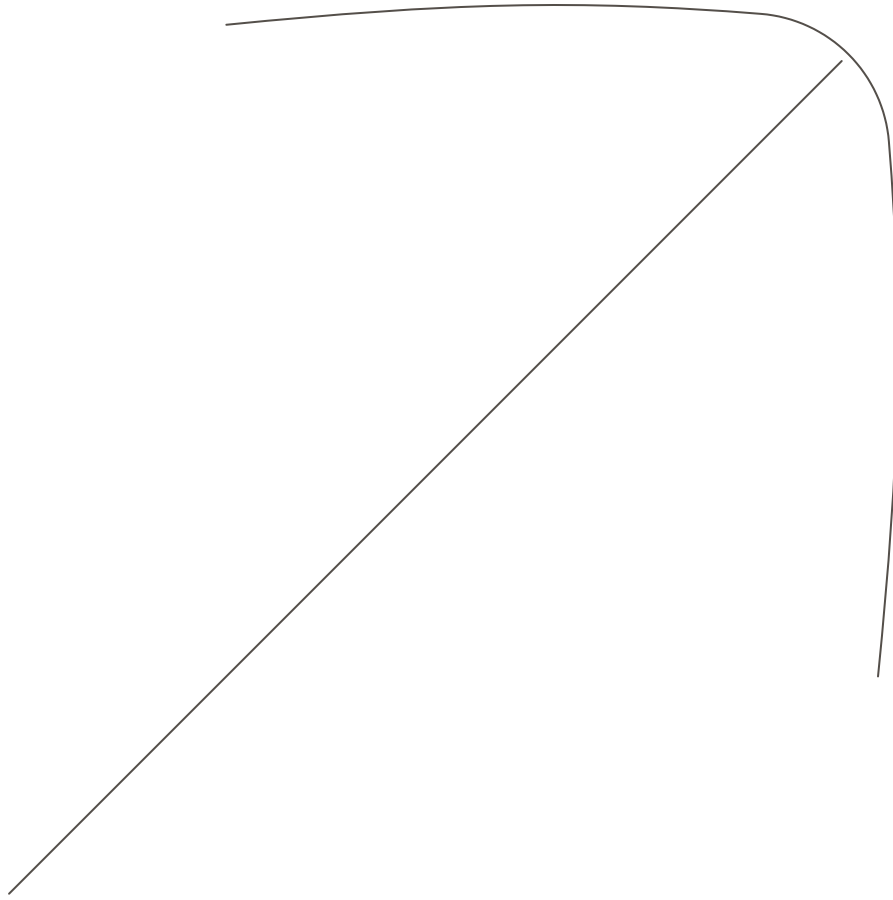
2. Empirical modeling and hybrid machine learning framework for nucleate pool boiling on microchannel structured surfaces (2025)



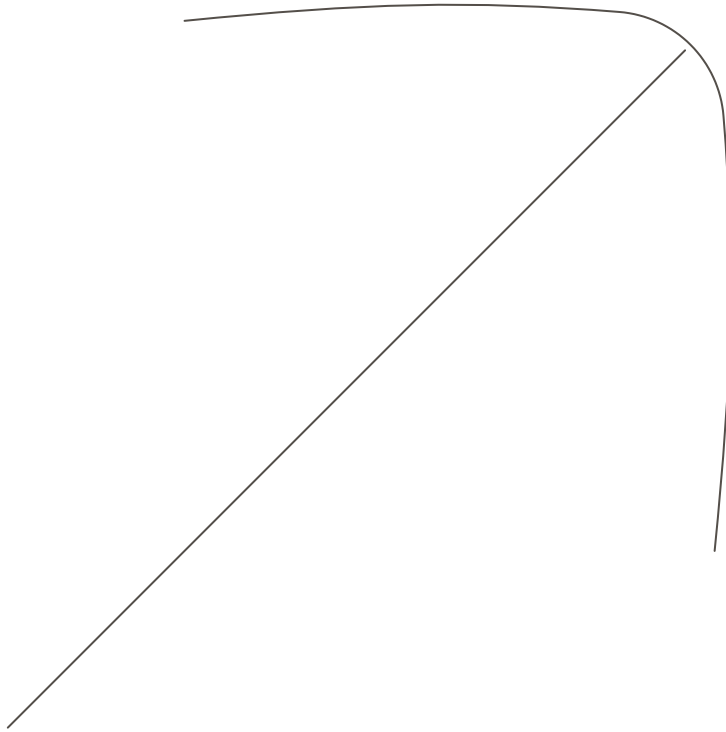
3. Physics-Based Hybrid Machine Learning for Critical Heat Flux Prediction with Uncertainty Quantification (2025)



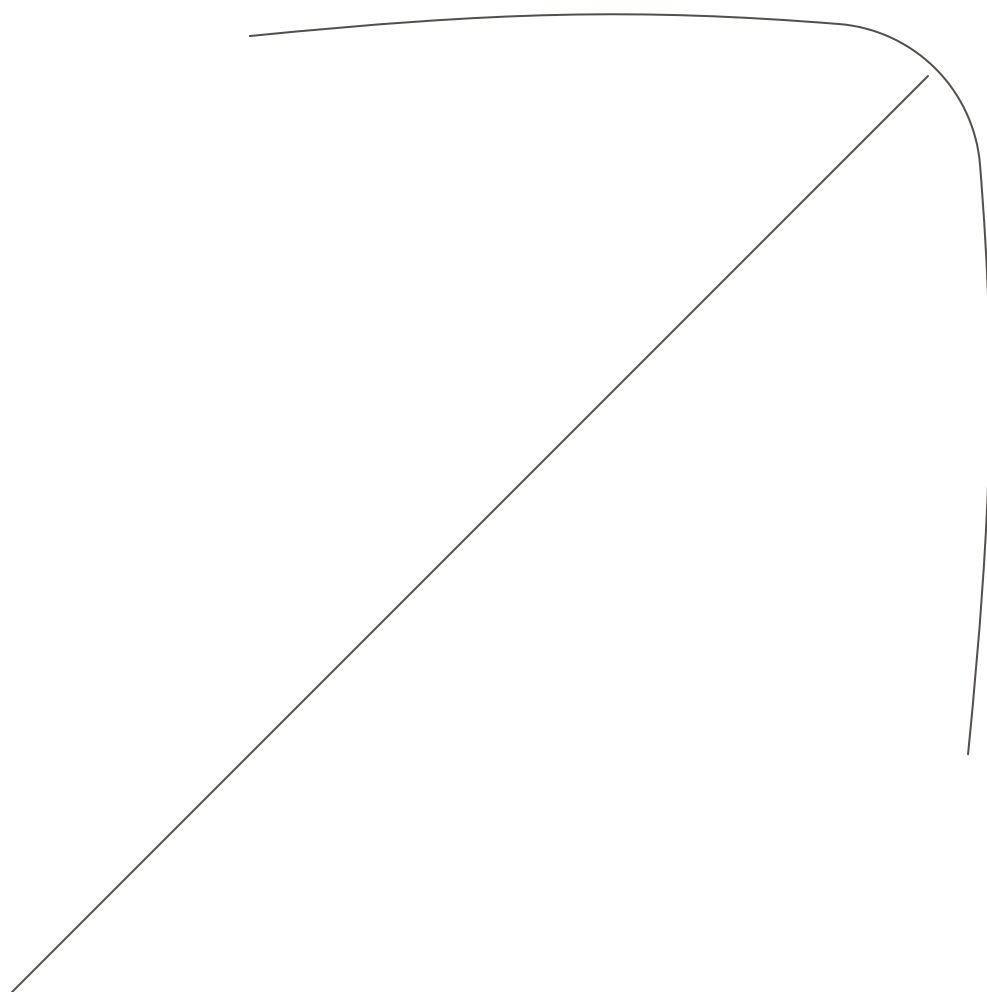
4. Hybrid analysis and modeling for next generation of digital twins (2021)



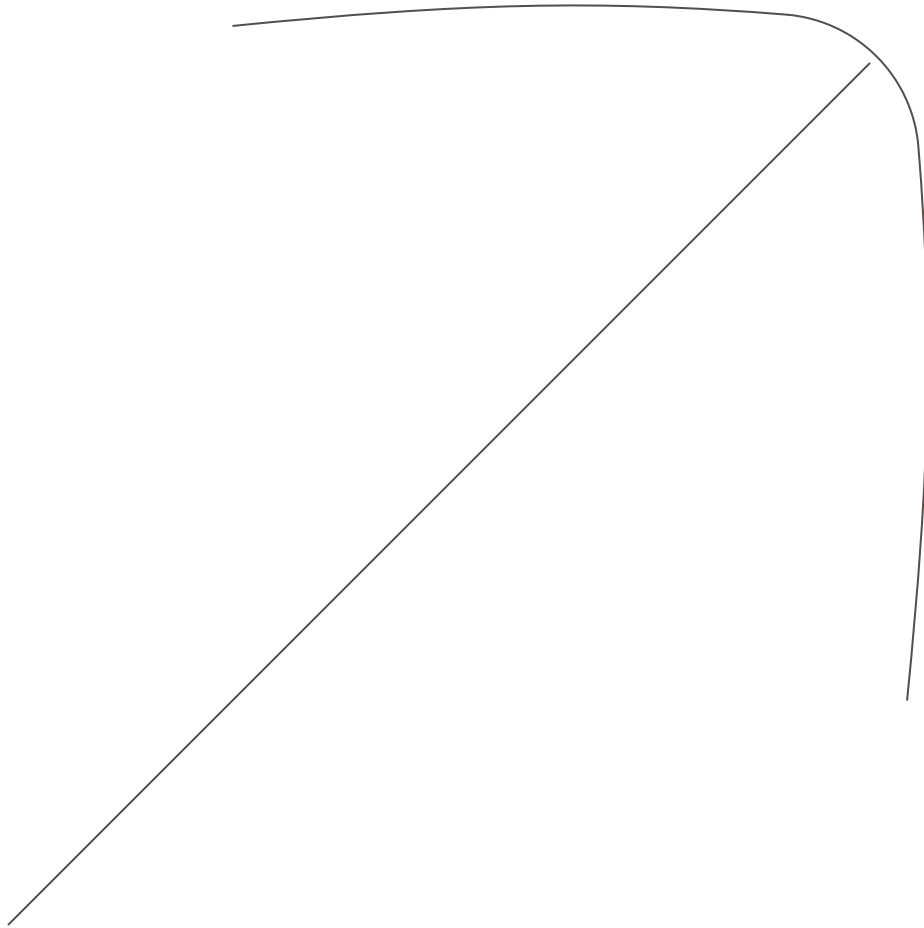
5. Model fusion with physics-guided machine learning (2021)



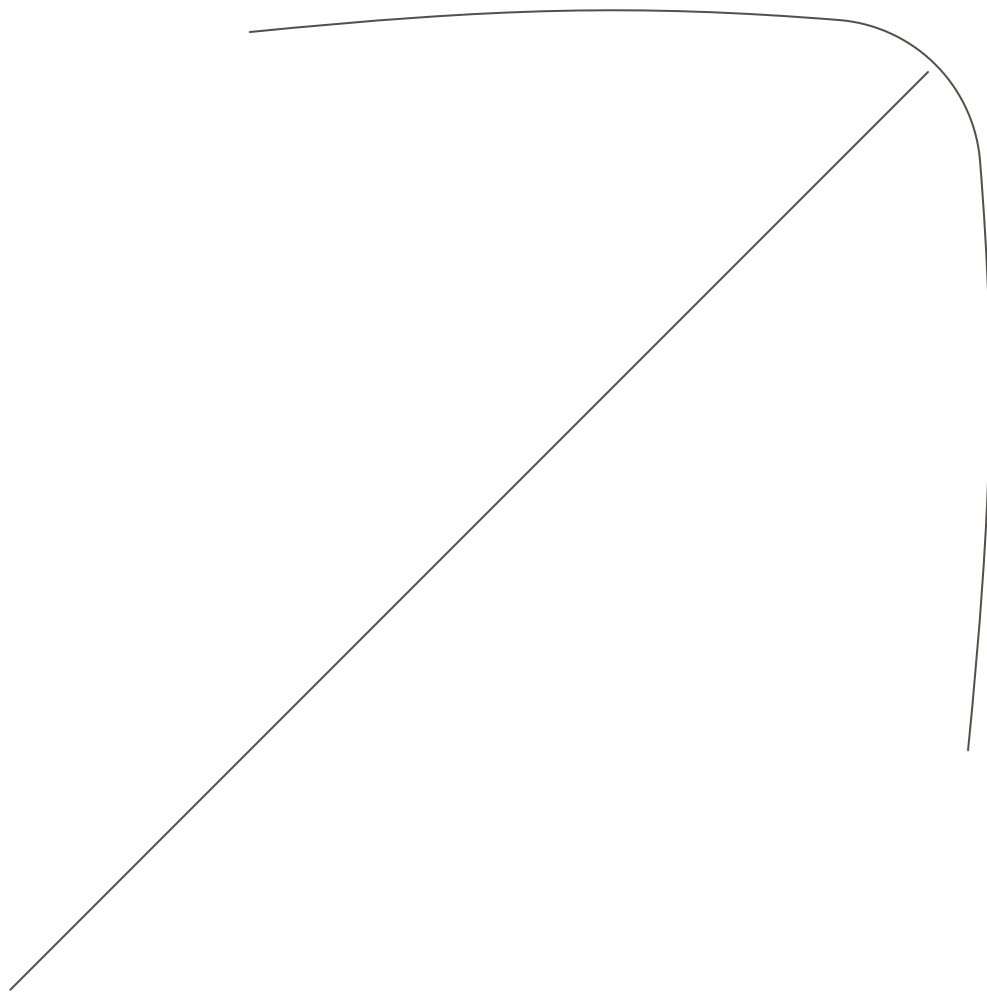
6. Hybrid Physics-Machine Learning Models for Quantitative Electron Diffraction Refinements (2025)



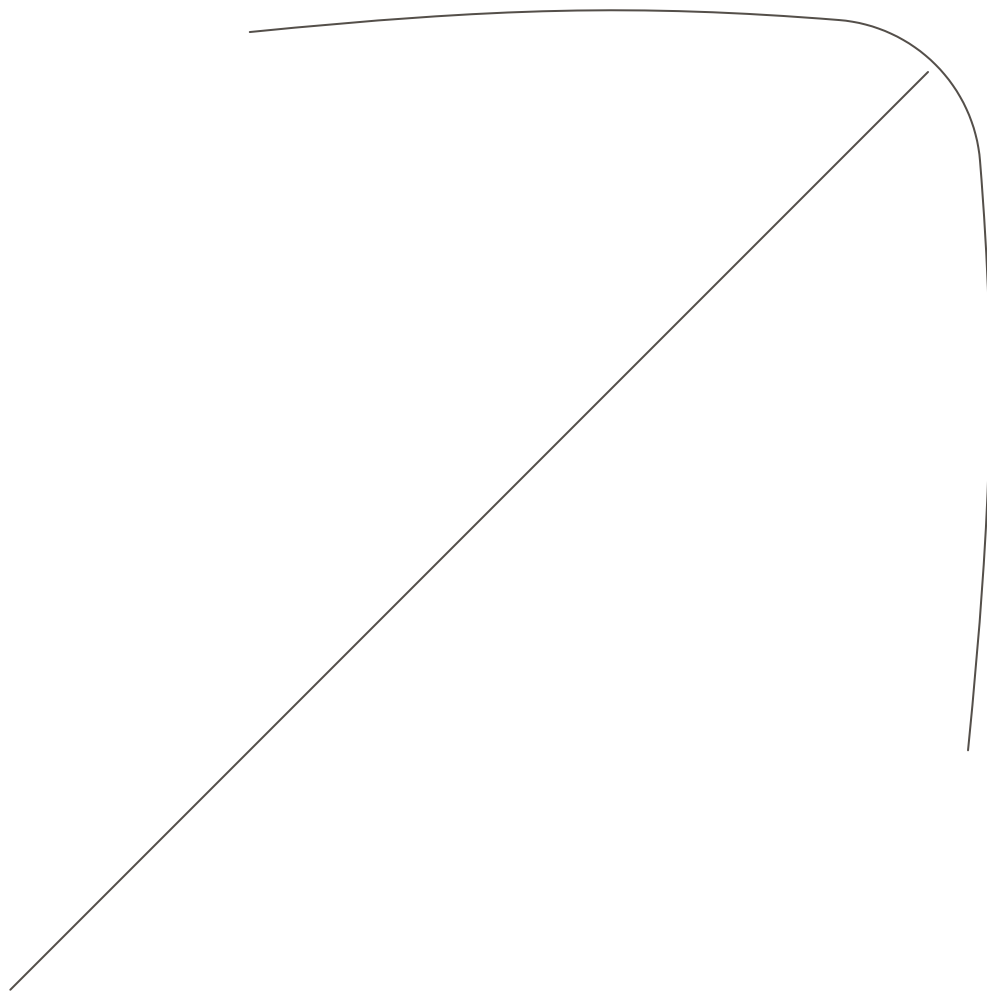
7. HYCO: Hybrid-Cooperative Learning for Data-Driven PDE Modeling (2025)



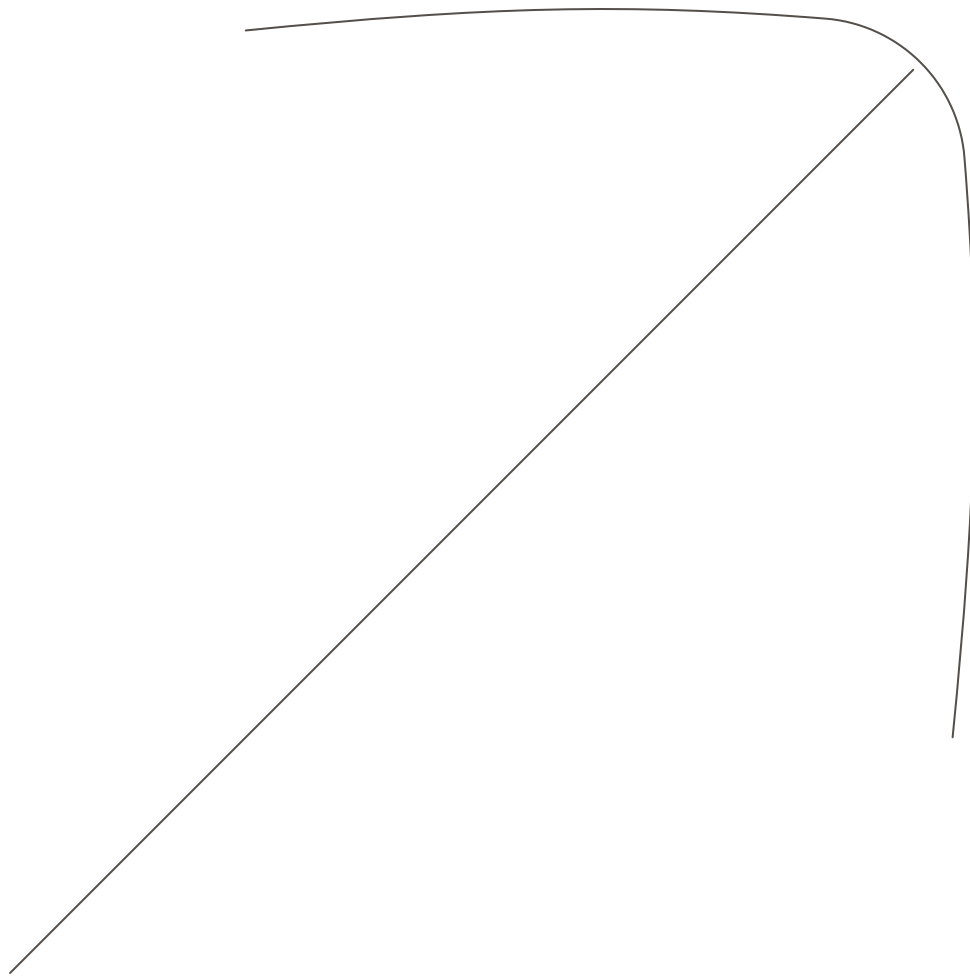
8. A hybrid data driven-physics constrained Gaussian process regression framework with deep kernel for uncertainty quantification (2022)



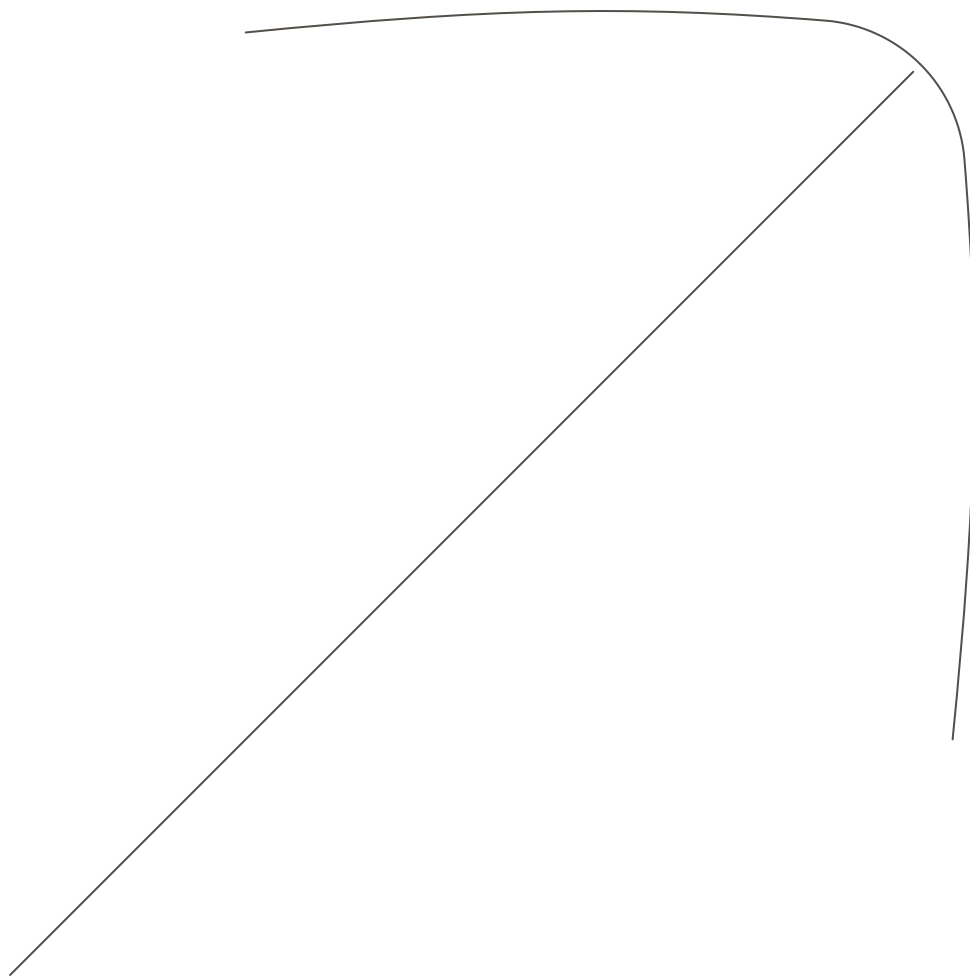
9. Hybrid physics-informed metabolic cybergenetics: process rates augmented with machine-learning surrogates informed by flux balance analysis (2024)



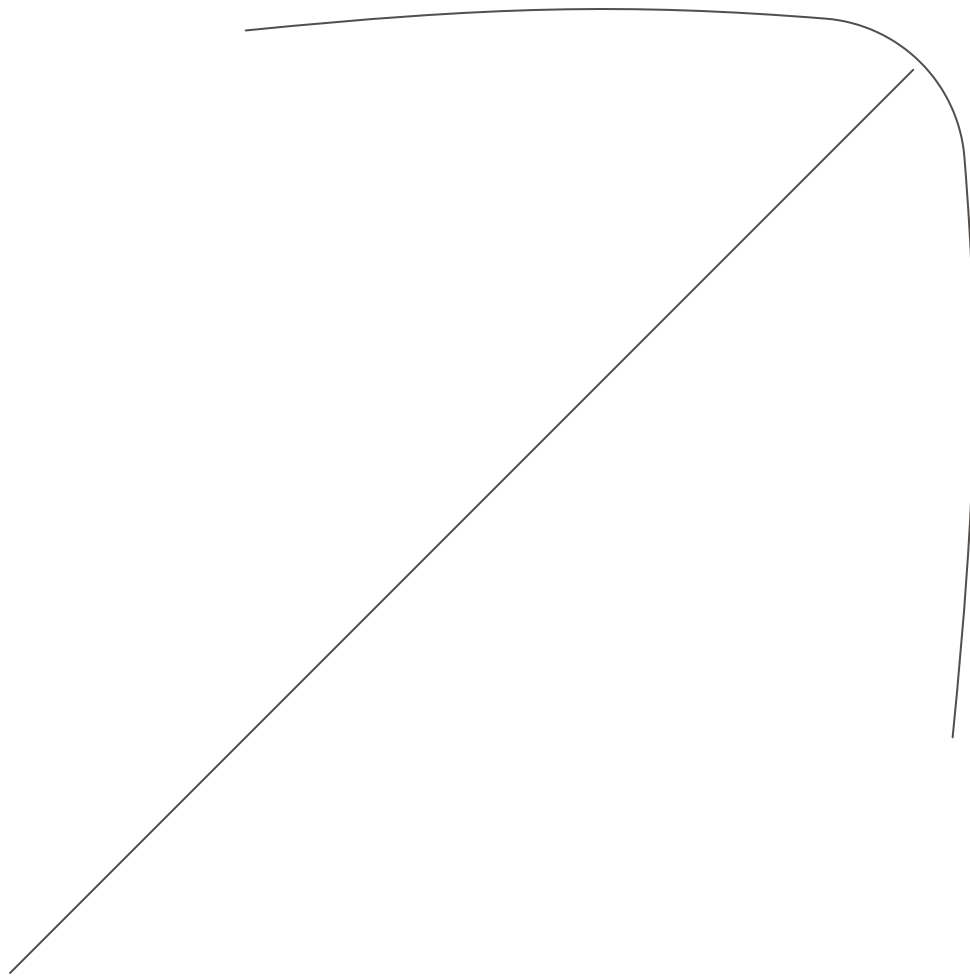
10. Hybrid Physics-ML Framework for Pan-Arctic Permafrost Infrastructure Risk at Record 2.9-Million Observation Scale (2025)



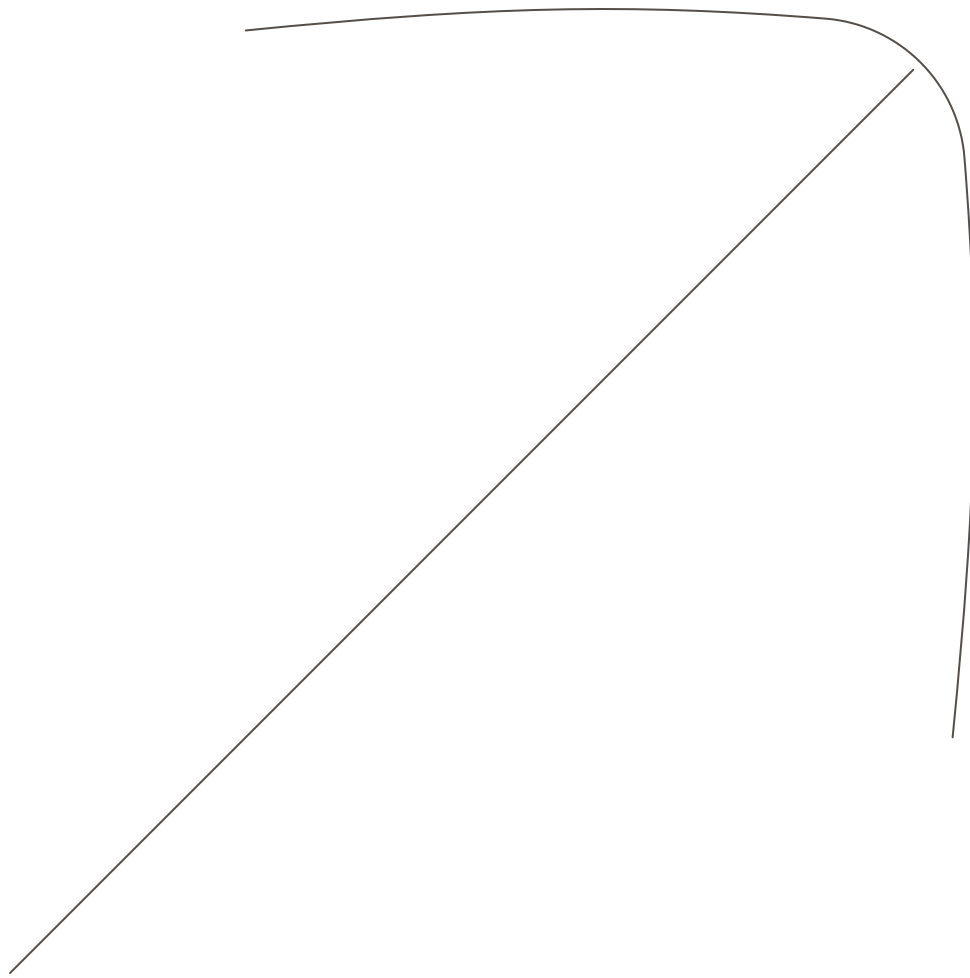
11. A nonintrusive hybrid neural-physics modeling of incomplete dynamical systems: Lorenz equations (2021)



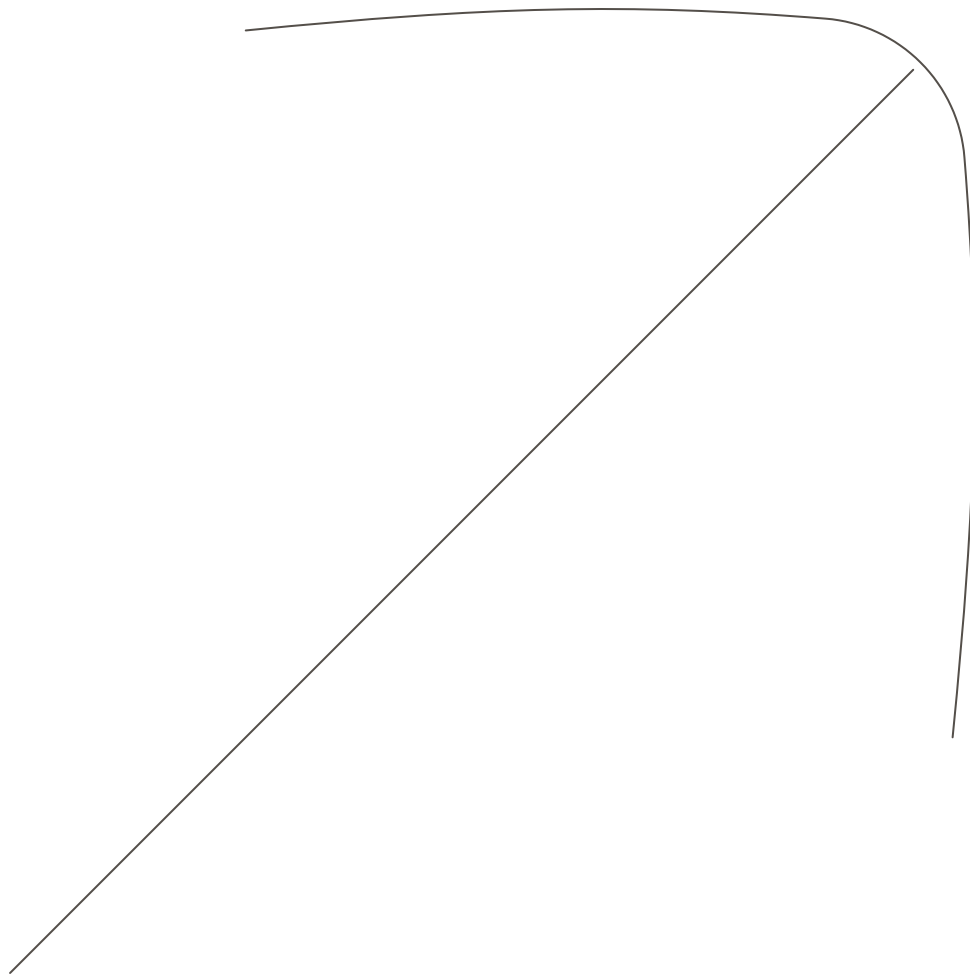
12. Joint Parameterization of Hybrid Physics-Based and Machine Learning Li-Ion Battery Model (2025)



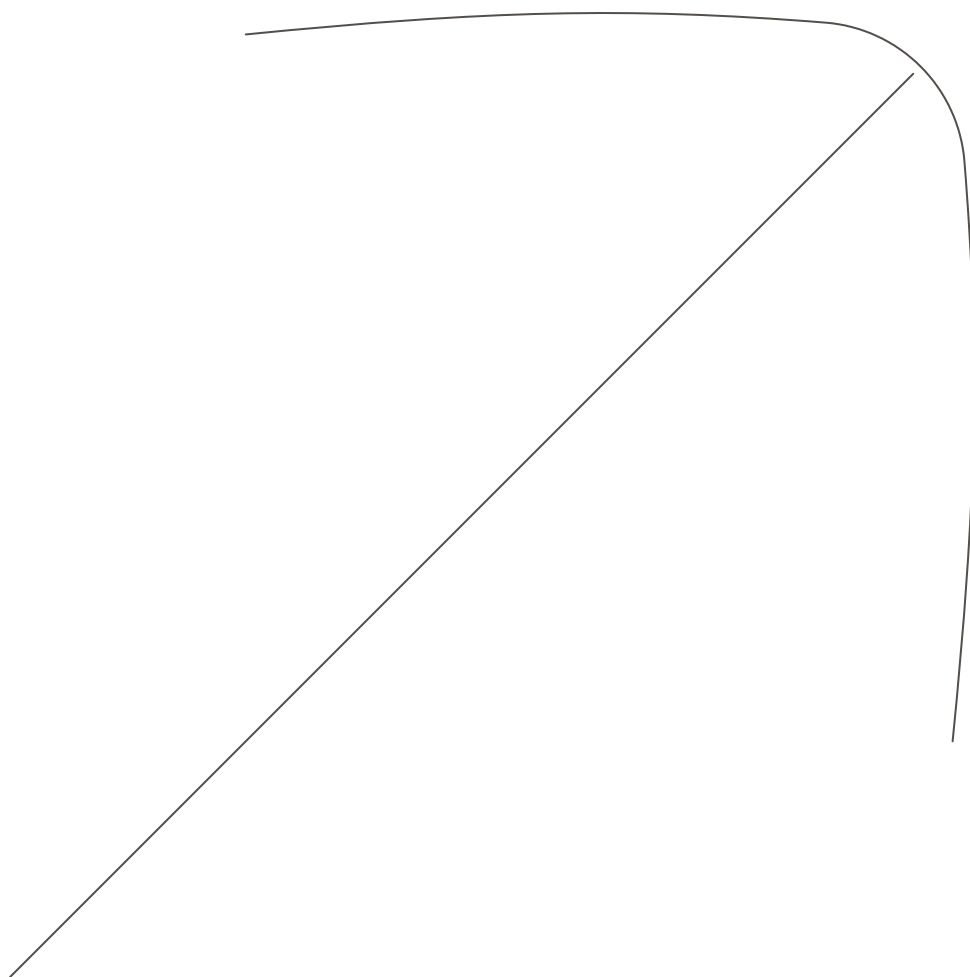
13. Hybrid Machine Learning and Physics-based Modelling of Pedestrian Pushing Behaviours at Bottlenecks (2024)



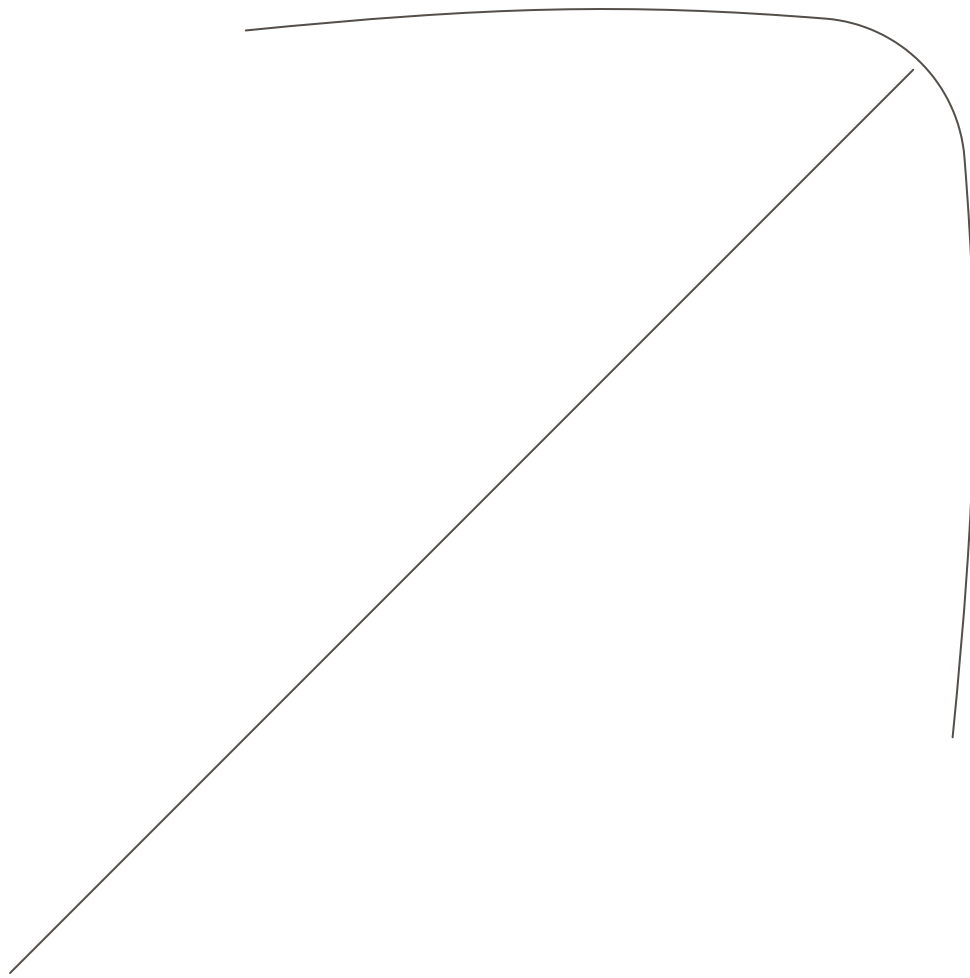
14. Hybrid Physics-ML Modeling for Marine Vehicle Maneuvering Motions in the Presence of Environmental Disturbances (2024)



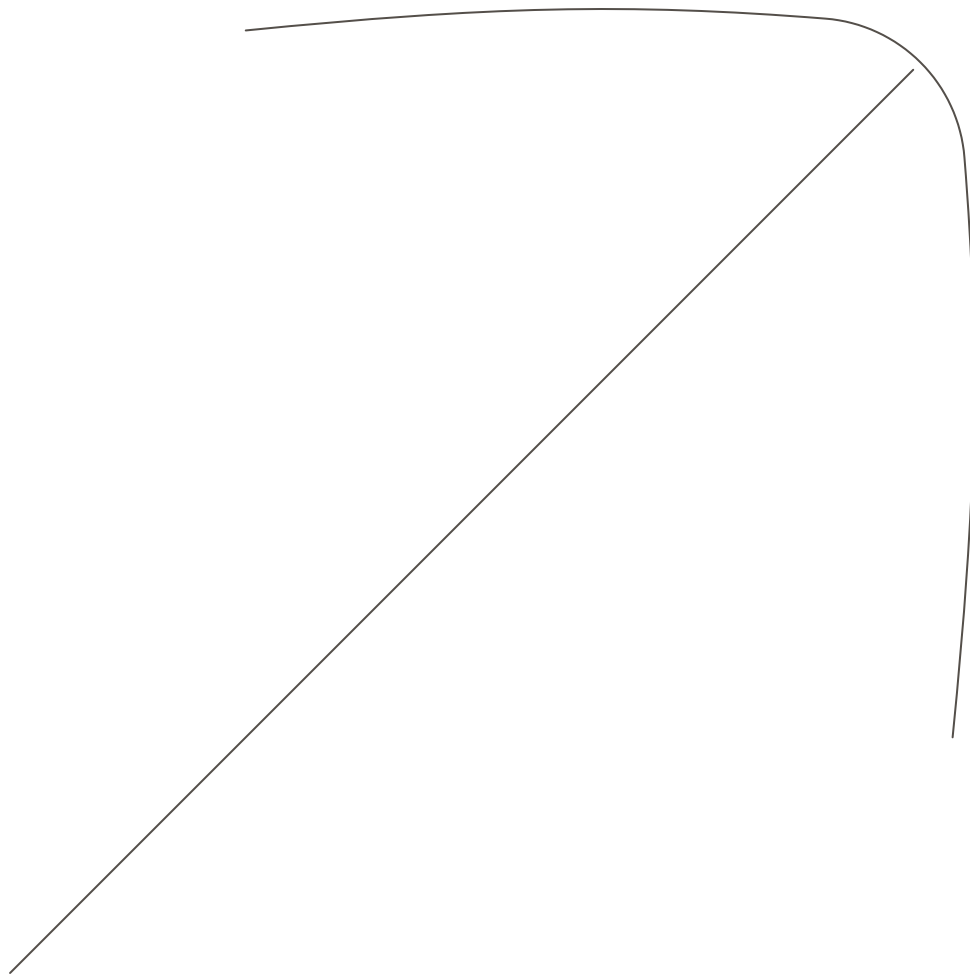
15. BridgeNet: A Hybrid, Physics-Informed Machine Learning Framework for Solving High-Dimensional Fokker-Planck Equations (2025)



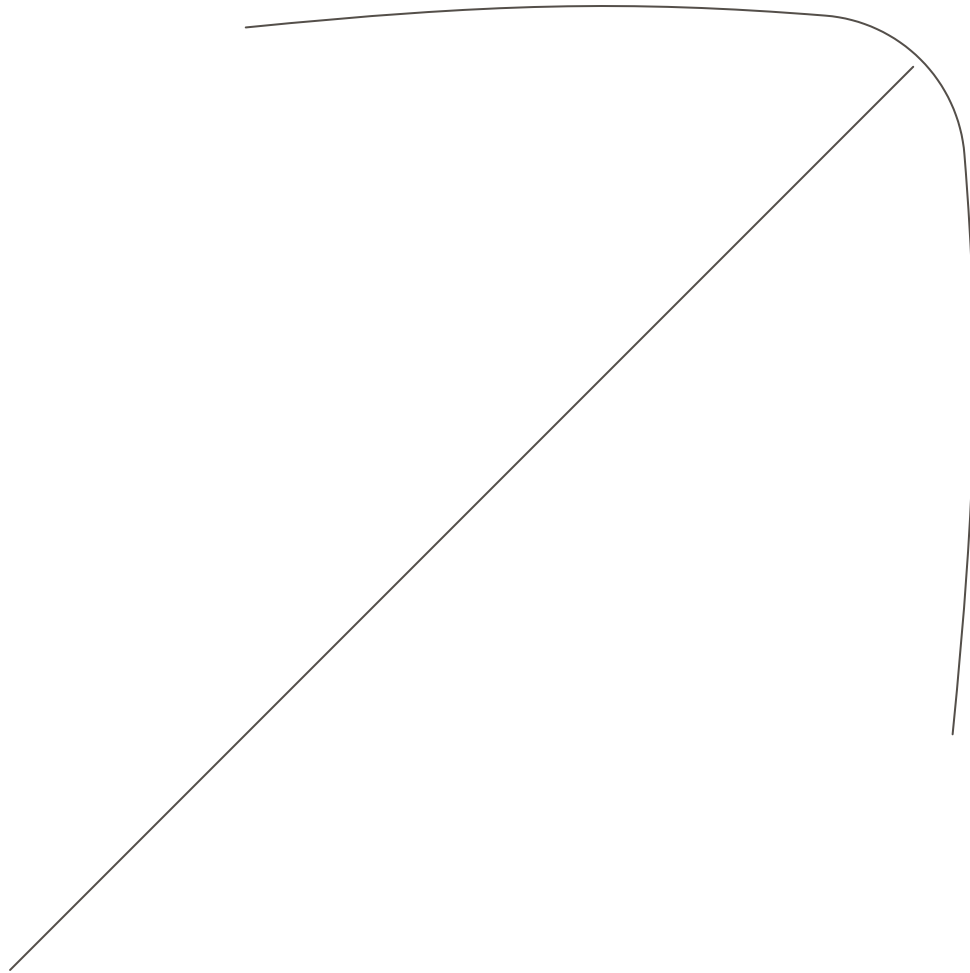
16. Data-Driven Computing Methods for Nonlinear Physics Systems with Geometric Constraints (2024)



17. Machine learning of evolving physics-based material models for multiscale solid mechanics (2023)



18. Hybrid Approaches for Black Hole Spin Estimation: From Classical Spectroscopy to Physics-Informed Machine Learning (2025)

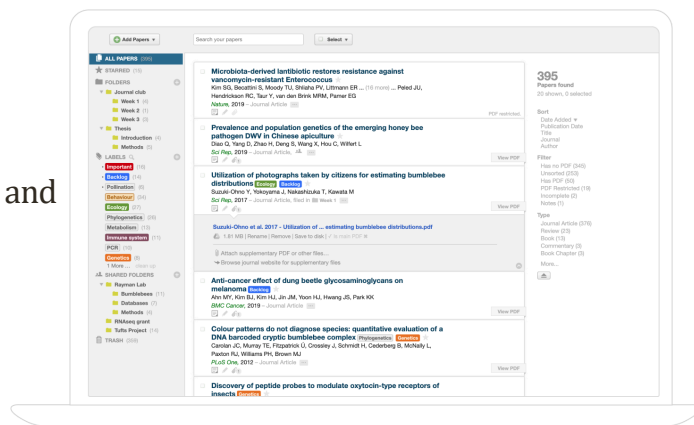


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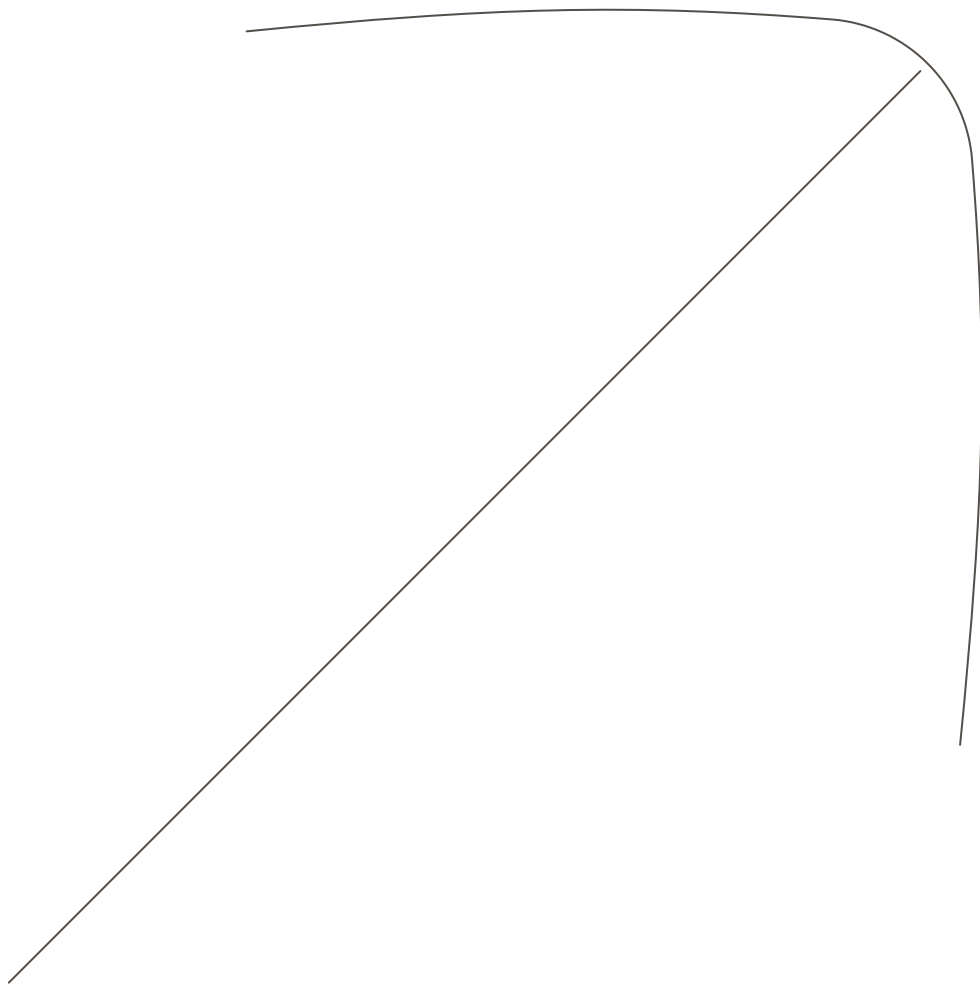
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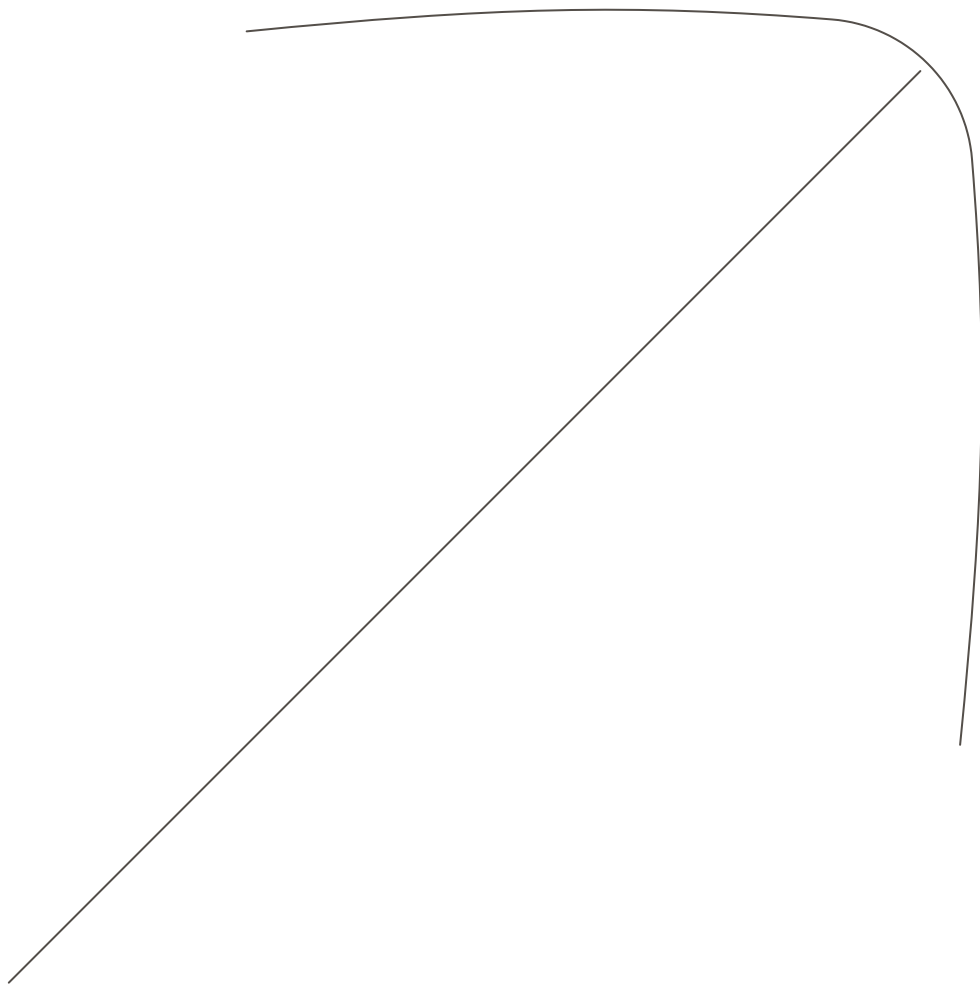


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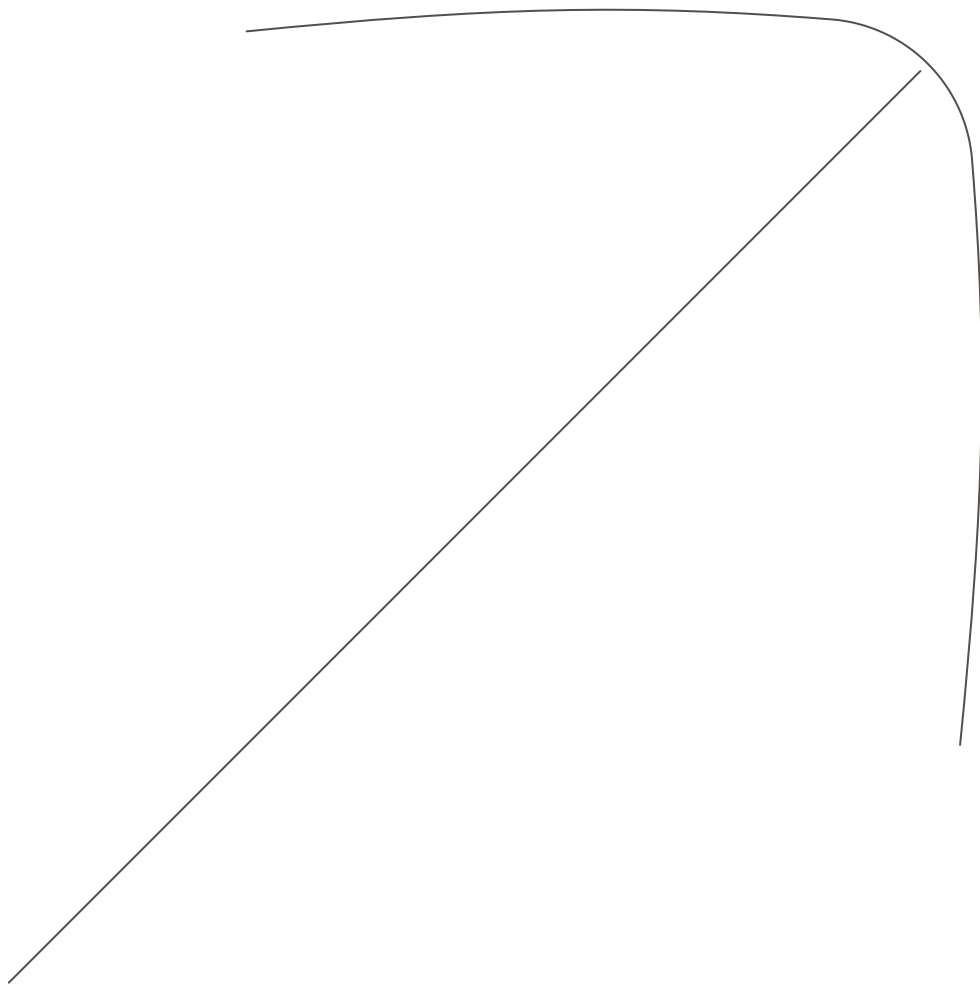
How do hybrid physics–ML frameworks combine mechanistic insights with data-driven methods for improved predictions?



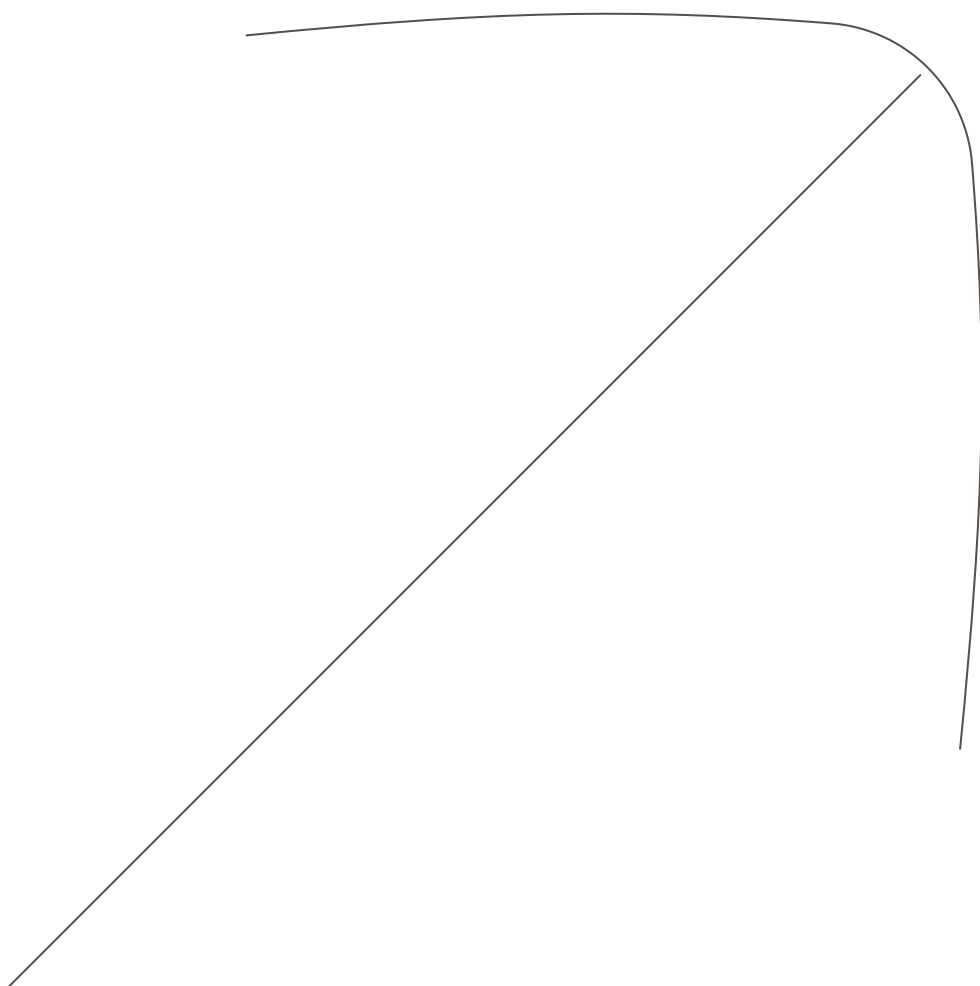
What are the key benefits of integrating physics-based models into machine learning architectures?



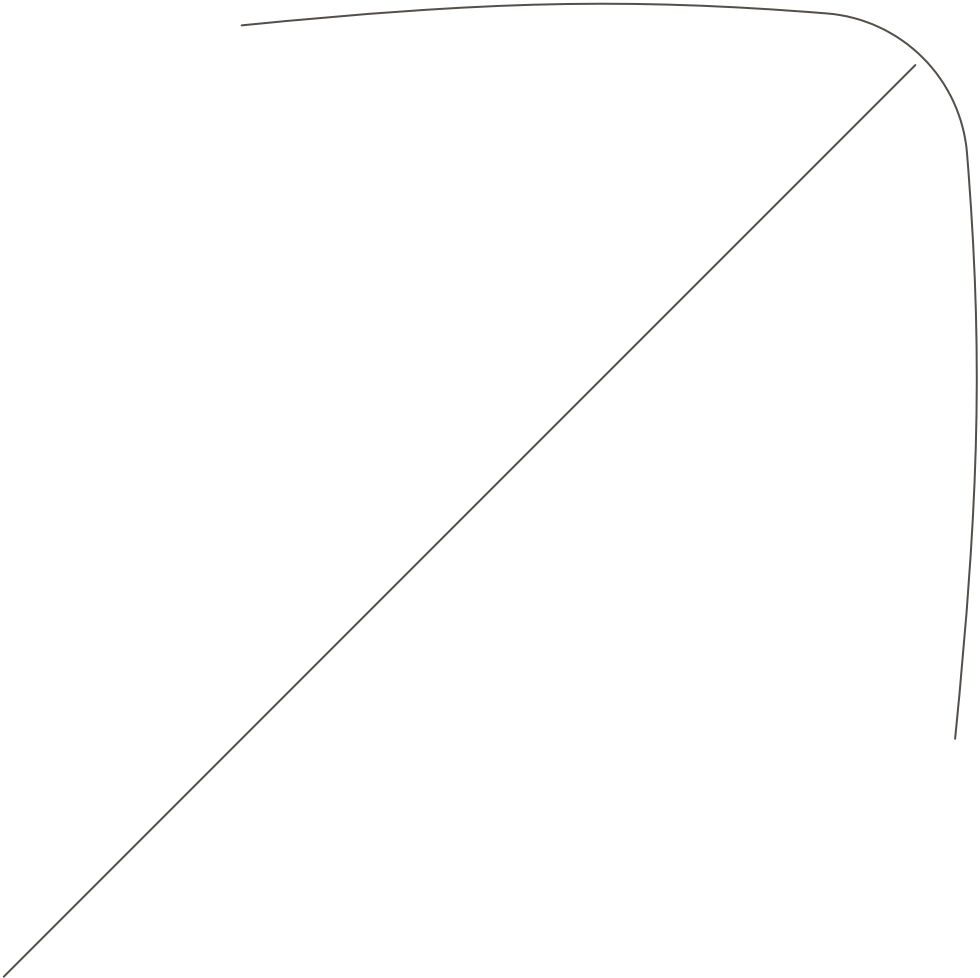
How can uncertainty quantification be enhanced in hybrid modeling approaches?



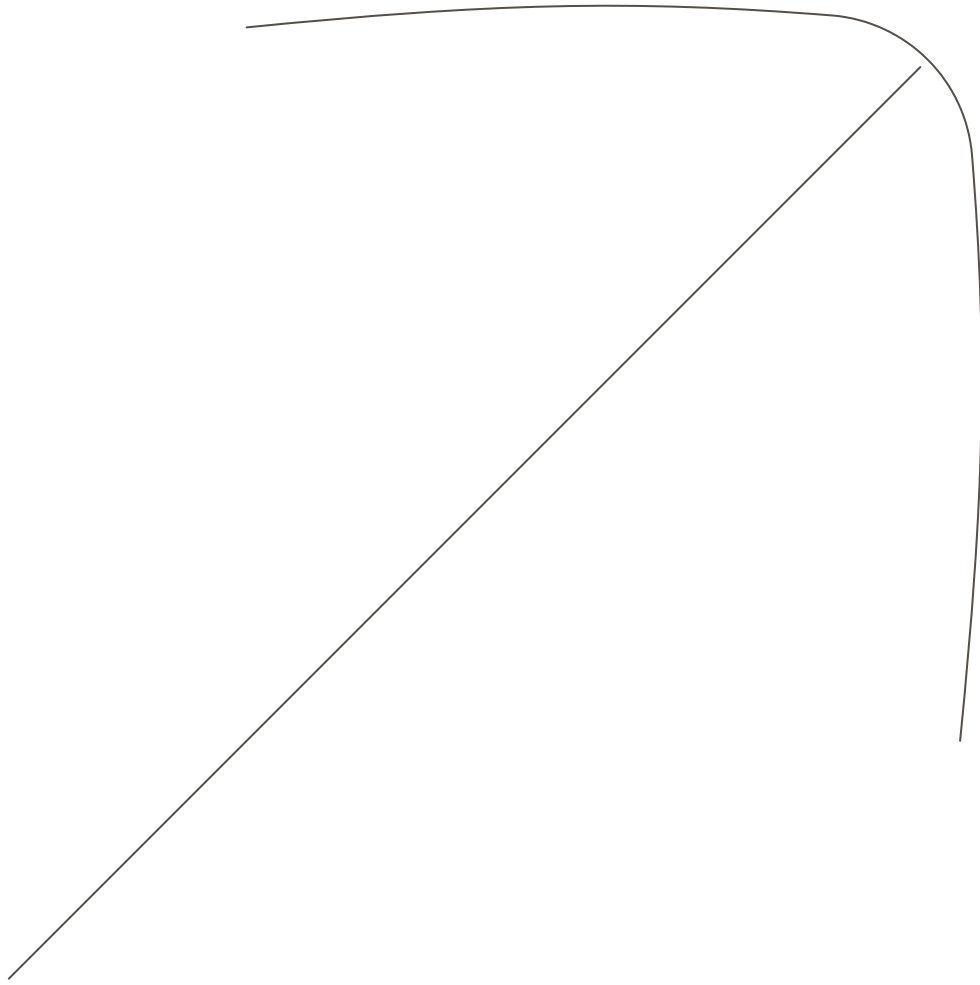
What challenges must be overcome when calibrating and validating hybrid physics-ML models?

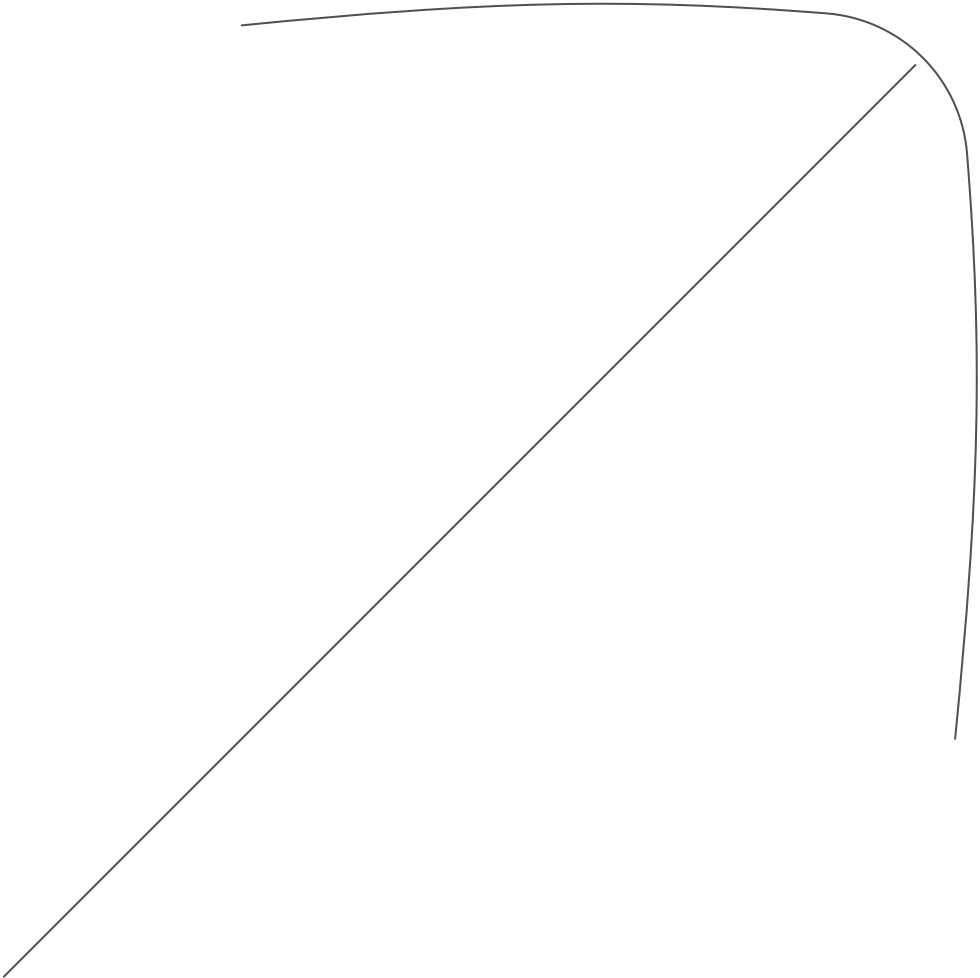


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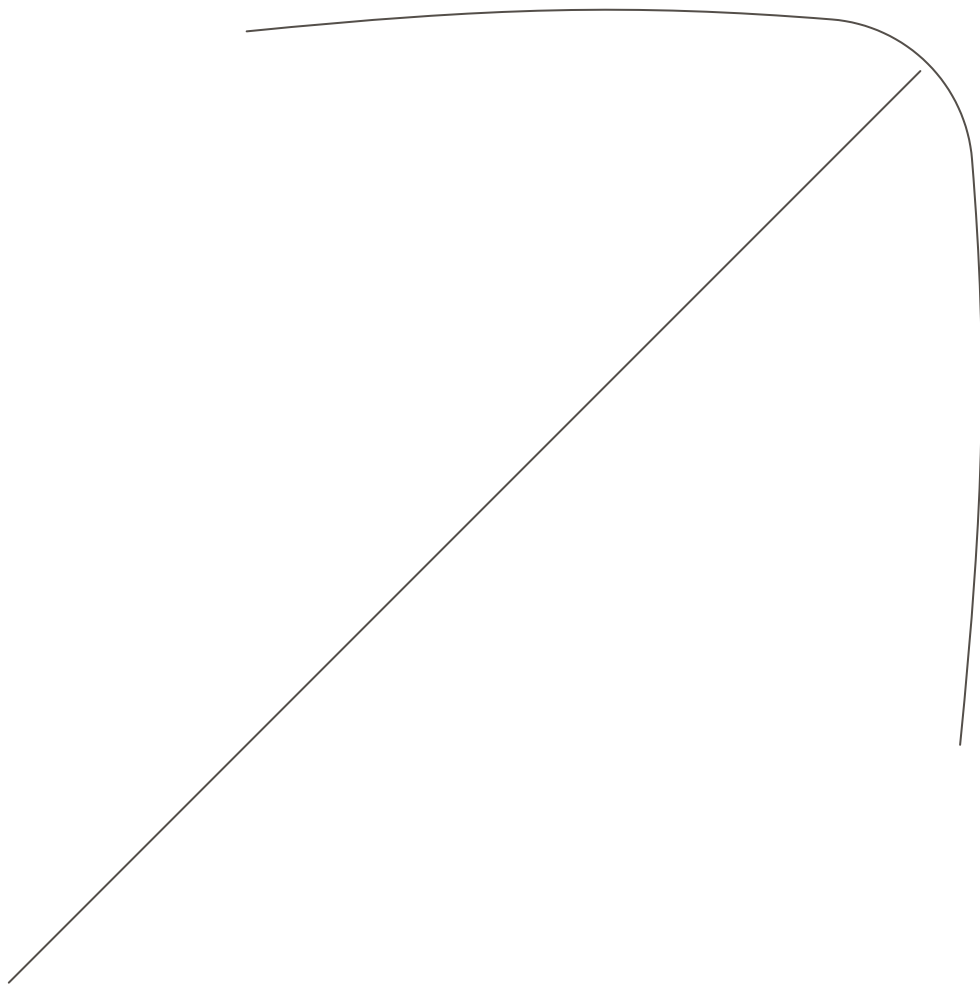


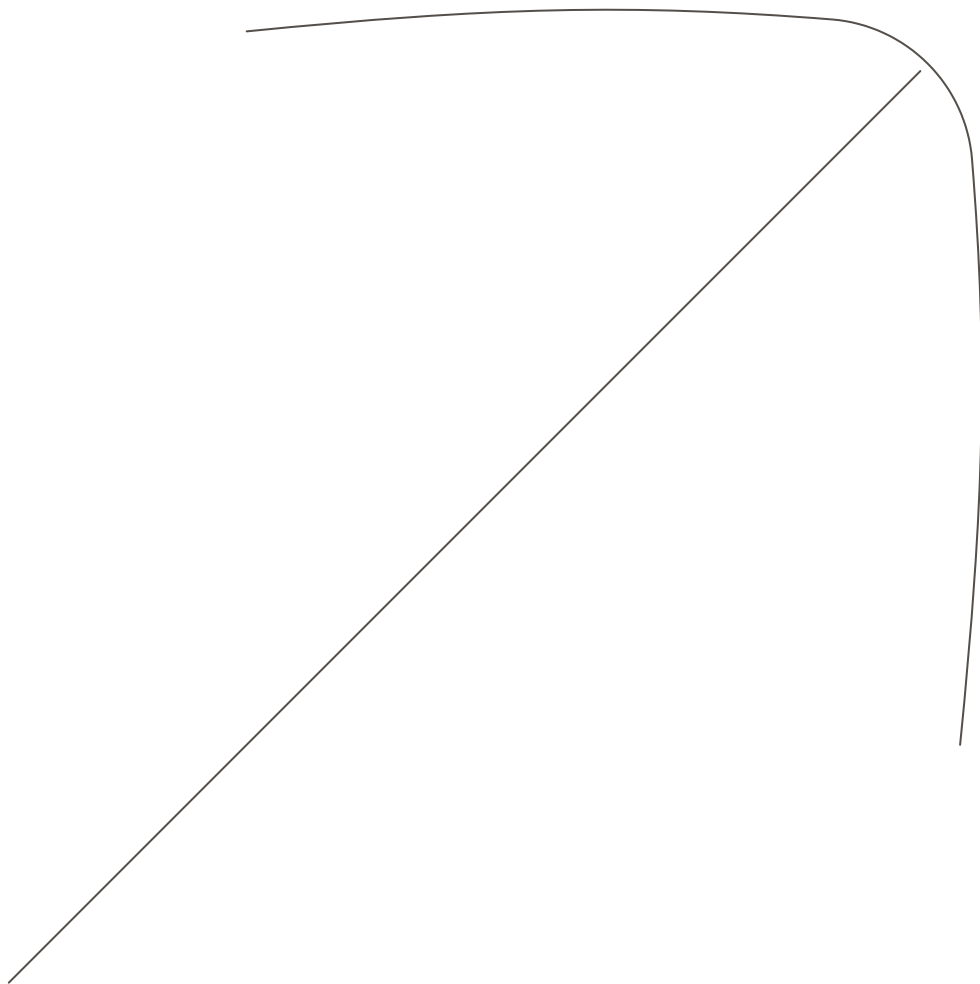
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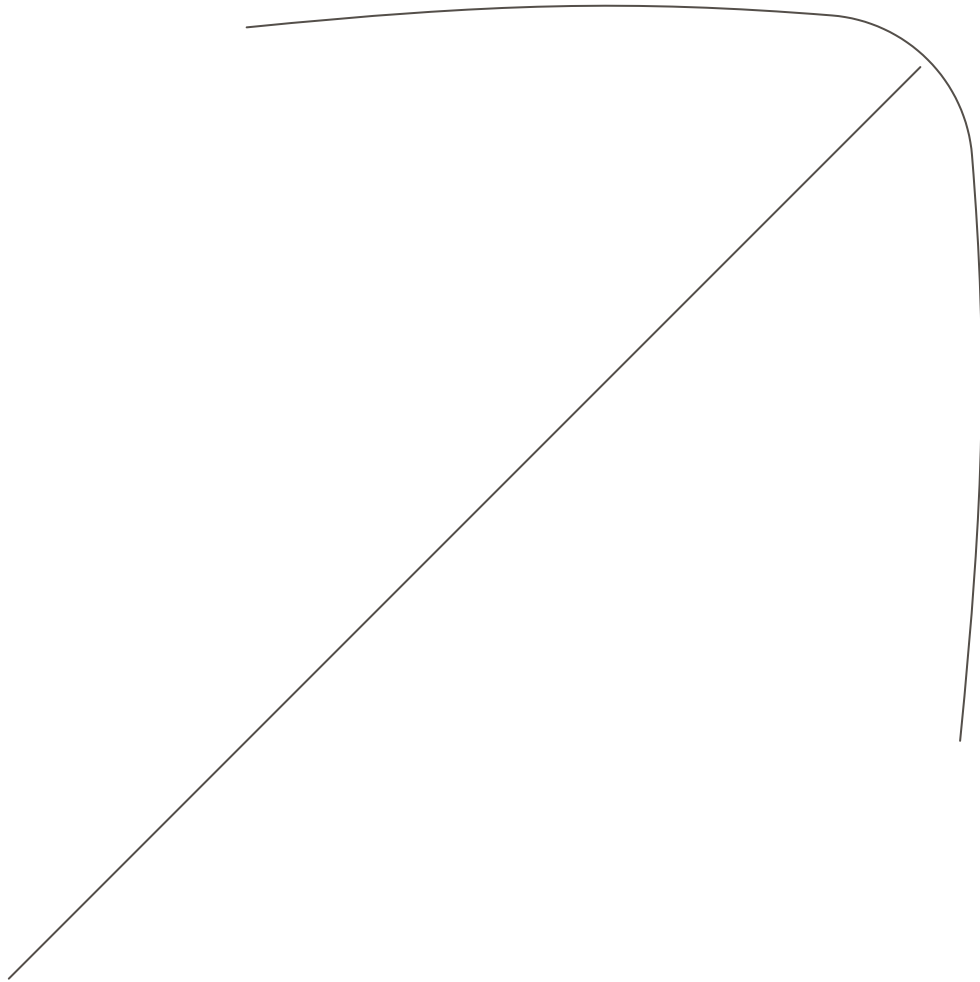




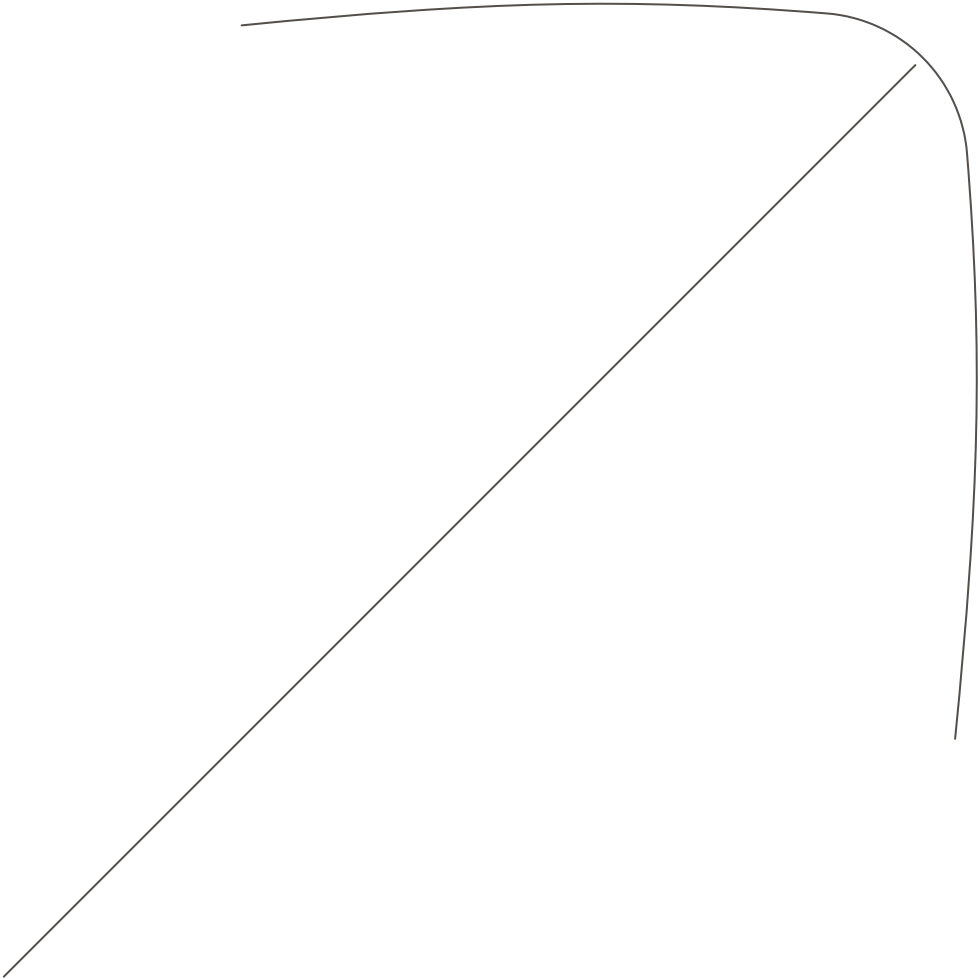
Hybrid Physics-ML Potentials

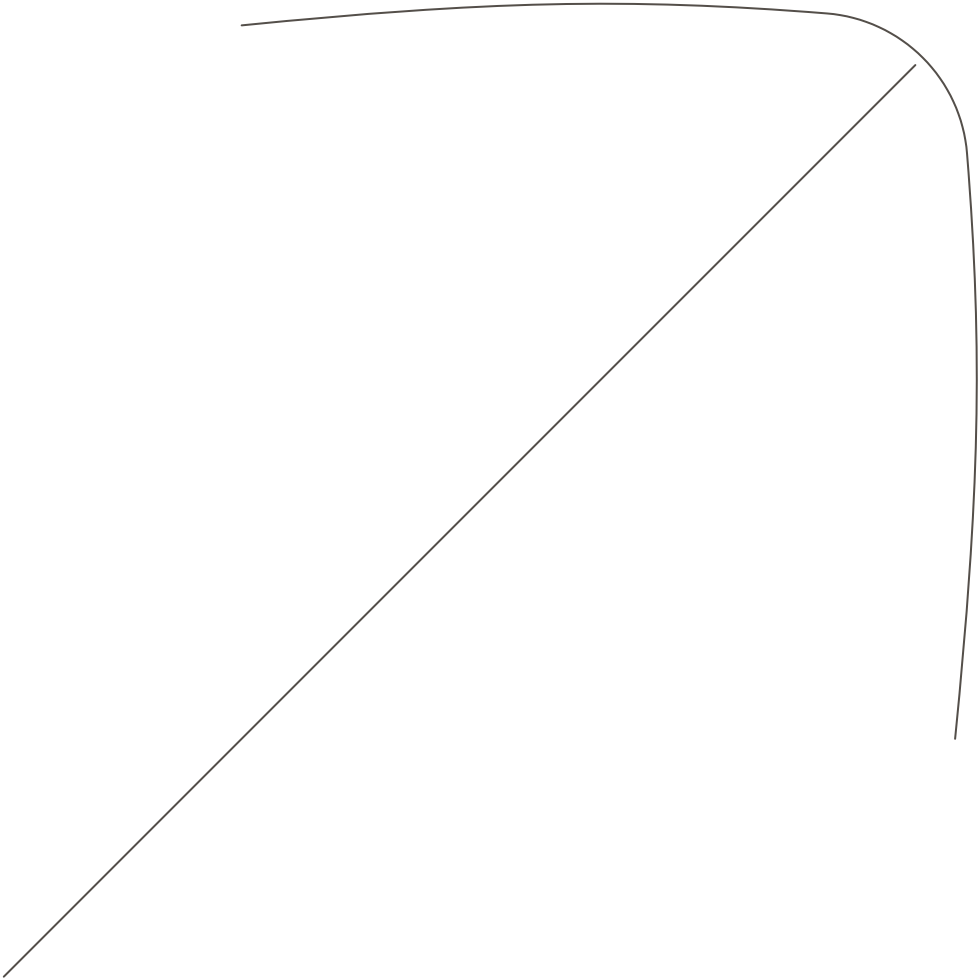




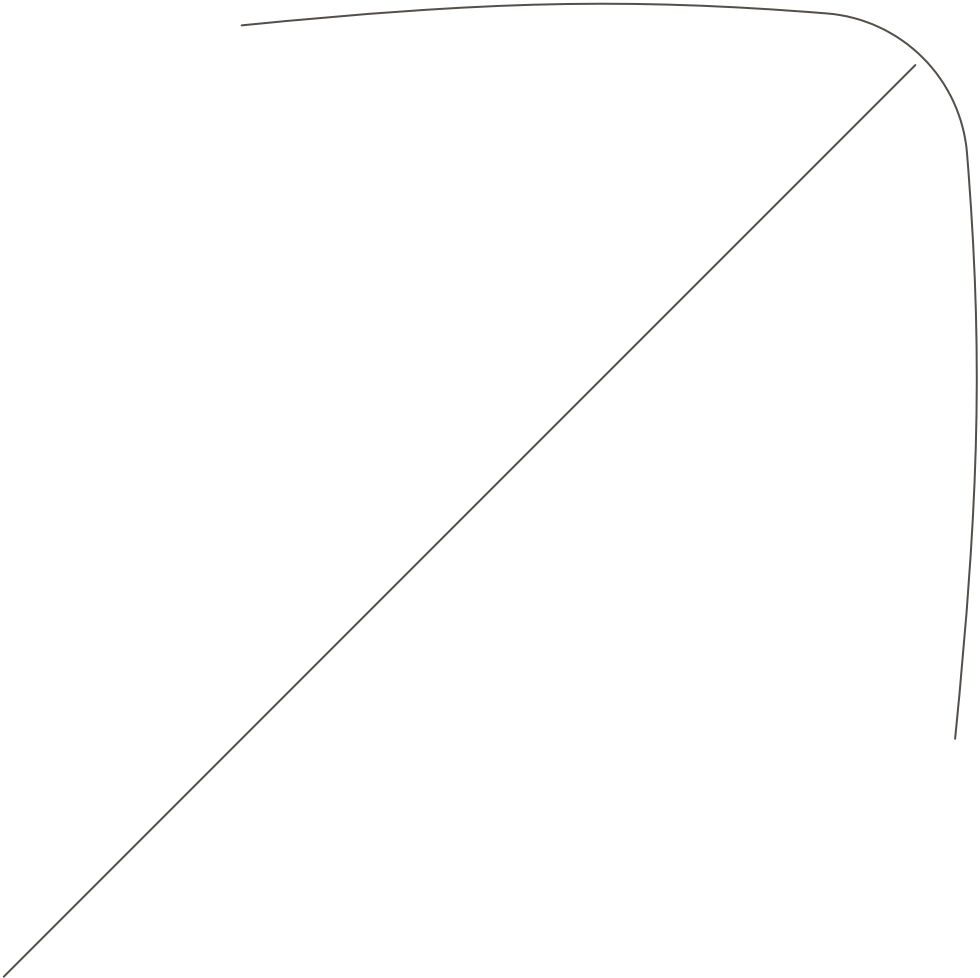


Hybrid Physics-ML Framework Overview

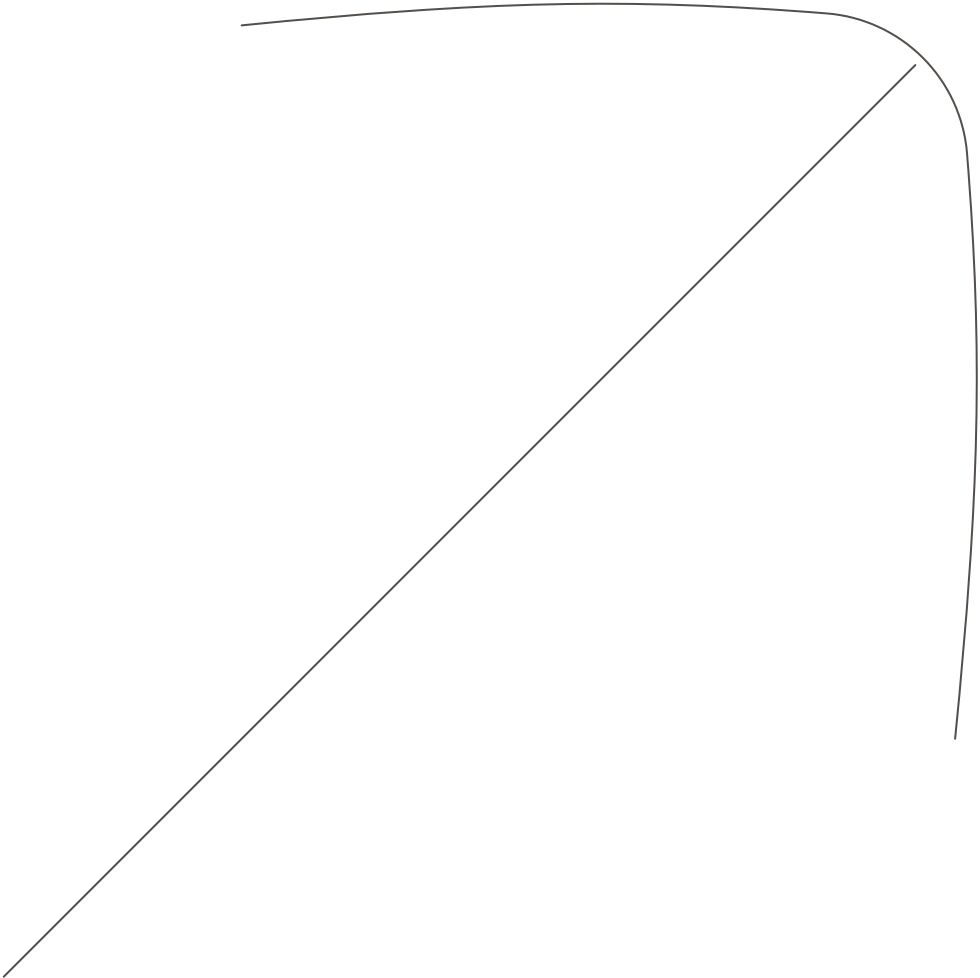




Hybrid Physics-ML Risk Framework



Multi-Fidelity Deep Learning Framework



ML-Enhanced Greybox Framework